Information Fusion 118 (2025) 102993



Contents lists available at ScienceDirect

Information Fusion

journal homepage: www.elsevier.com/locate/inffus



Full length article



Advances in DeepFake detection algorithms: Exploring fusion techniques in single and multi-modal approach

Ashish Kumar [a], Divya Singh [a], Rachna Jain [b], Deepak Kumar Jain [c,d,*], Chenquan Gan [e], Xudong Zhao [c]

[a] School of Computer Science Engineering and Technology, Bennett University, Greater Noida, U.P., India
[b] Bhagwan Parshuram Institute of Technology, New Delhi, India
[c] Key Laboratory of Intelligent Control and Optimization for Industrial Equipment of Ministry of Education, Dalian University of Technology, Dalian 116024, China
[d] Symbiosis Institute of Technology, Symbiosis International University, Pune 412115, India
[e] School of Cyber Security and Information Law, Chongqing University of Posts and Telecommunications, Chongqing 400065, China

ARTICLE INFO

Keywords:
DeepFake
Artificial intelligence
Generative adversarial network
Fusion algorithms
Transformer
Detection

ABSTRACT

In recent years, generative artificial intelligence has gained momentum and created extremely realistic synthetic multimedia content that can spread misinformation and mislead society. Deepfake detection is a technique consisting of frameworks, algorithms and approaches to predict manipulated contents namely, image, audio and video. To this end, we have analyzed and explored various deepfake detection frameworks by categorizing them as single-modal or multi-modal approaches. To provide better understanding and clarity, single-modal approaches are further categorized as conventional and advanced techniques. Conventional techniques extract complementary handcrafted features and classify them using machine-learning-based algorithms. On the other hand, advanced techniques adopt deep learning and hybrid algorithms to detect deepfakes. Multi-modal techniques utilize a mixture of two or more modalities for feature extraction and fuse them to obtain the final classification scores. These techniques are also categorized either as deep learning or hybrid techniques. The complementary features, multiple modalities, and deep learning models are fused adaptively using score-level or feature-level fusion. The advantages, features, practical applications, and limitations under each category are highlighted to address the challenges and determine future trends to counter deepfakes. In addition, recommendations are also elaborated to evaluate the potential of artificial intelligence in deepfake detection for providing a safer and more reliable digital world.

1. Introduction

With the advancement in digital media, the contamination and tampering in multimedia content such as images, videos, and audio have been a foremost concern nowadays [1]. The detection of alteration is paramount at an early stage to prevent the dissemination of fake and incorrect content to the general people with the use of social platforms such as Facebook, Instagram, and many other [2,3]. The fraudster's content in the form of audio, video, and images are created with the help of artificial intelligence (AI)-based technology such as GANs (Generative adversarial networks) by altering the real content with false data and are popularly known as Deepfakes [4]. DeepFakes are technologically advanced content that superimposes fake and inappropriate details on the original content to generate false authentic information.

Deepfakes are artificially synthesized contents that require a large amount of training data for their generation [5]. Deepfakes generate fake and inappropriate images and videos by exchanging the faces of well-known personalities such as celebrities, and superstars. Deepfakes can produce fake pornographic material in the form of realistic videos and images using a person's face without his/her permission [6]. Also, the speech and voice content of politicians and world leaders can be modified to spread fake news and misinformation, impacting their visibility and political careers. Deepfakes can create confusion during military movements, misguide forensic investigations, tamper the satellite images, and cause religious tension too [7–10]. Hence, these contents not only deceive the individual people but are also a threat to society and nations worldwide. The detection of deepfakes is

* Corresponding author at: Key Laboratory of Intelligent Control and Optimization for Industrial Equipment of Ministry of Education, Dalian University of Technology, Dalian 116024, China.
E-mail addresses: ashish.gupta14d@gmail.com (A. Kumar), divya.singh@bennett.edu.in (D. Singh), rachnabtec@gmail.com (R. Jain), dkj@ieee.org (D.K. Jain), gcq2010cqu@163.com (C. Gan), xudongzhao@dlut.edu.cn (X. Zhao).

https://doi.org/10.1016/j.inffus.2025.102993
Received 10 October 2024; Received in revised form 5 January 2025; Accepted 28 January 2025
Available online 5 February 2025
1566-2535/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

essential to prevent their ill effect from penetrating the human being.

Deepfakes are so realistic that they are difficult to identify with the naked eye or ears. It requires innovative and latest technology to differentiate the deepfakes from the original content. Recently AI-based algorithms have provided superior results in object detection in the presence of varying environmental conditions [11–15]. Also, these algorithms can predict critical diseases such as cancer and tumors at an early stage with enhanced accuracy [16–19]. These algorithms have a wide range of applications in detecting deepfakes in multimedia content such as audio, video, and images [20]. AI-based algorithms can utilize either the machine-learning (ML) approach [21–23] or the Deep learning (DL) approach [4,24–27] for detecting deepfakes. ML algorithms such as random forest (RF) [23,28], decision tree (DT) [28,29], XGBoost [28,30], support vector machine (SVM) [31,32], and many others [8,33] are utilized for classifying the content as real or fake. On the other hand, DL-models such as convolutional neural network (CNN) [34,35], ResNet [36], and EfficientNet [37] are exploited for learning robust features for deepfake detection. Also, transformers such as vision transformer (ViT) [38,39], Swim transformer [40], and student-teacher network [41] comprising encoder and decoder stages are efficient in deepfake detection. These models are data-hungry, computationally complex, and require high training time for feature extraction.

To address the limitations of these advanced algorithms, handcrafted features such as color, texture, and histogram of gradient (HOG) are extracted from image frames for detecting deepfakes [42–44]. Apart from these features, facial points, frame level features, set-level features, and feature distillation are also recommended for robust feature extraction [45–48]. These features are easier to extract and process. However, detection based on single features is not efficient enough and prone to produce false outcomes. Hence, multi-modal deepfake detection models that can consider multiple modalities are demonstrated in this field [41,49,50]. The multiple modalities are extracted which can compensate for each other during the detection process. Multi-modal information is extracted by combining multiple inputs of multimedia data such as audio, video, and images for developing a robust deepfake detection model. These models learn features from different modalities in different dimensions. The features were integrated using DL models or ensembled approaches [51–53].

1.1. Comparison with previous surveys

Deepfake detection is one of the challenging and evolving domains of computer vision. A lot of survey papers have been published in the recent past with a special focus on deepfakes images or videos. A few had reviewed audio with images for deepfake detection in practical scenarios. In 2022, Rana et al. [54] reviewed and summarized the key algorithms in deepfake detection in four different categories namely, DL-based, ML-based, statistical, and blockchain-based algorithms from 2018 to 2020. However, conventional techniques were not studied and discussed. The authors comprehensively analyzed the deepfake generation methods, and DL-based detection techniques along with a discussion of available datasets [2]. The authors also discussed the role of body motion reenactment and its technicality in generating deepfake to explore the new directions in the domain. Similarly, authors analyzed and summarized the various DL-based deepfake creation and detection methods [55]. The discussion majorly described the technicalities of detecting deepfakes using DL methods and the available public datasets only. However, studies with multi-modal information with audio as input features were limitedly included in the discussion. Patel et al. [1] studied the ML/DL methods for deepfake creation and the advanced encoder/decoder-based detection methods. The benefits and limitations of each method were elaborated and discussed. However, the study failed to examine the conventional and multi-modal deepfake detection approaches. On the other hand, authors have reviewed the potential of AI for automatic generation and detection of deepfakes [20]. ML/DL based methods, facial expressions, and face reenactment were elaborated for deepfakes generation along with the associated implementation changes. A lot of deepfake data is created and is available publicly, which not only stains one's reputation but can also spread misinformation. Similarly, numerous deepfake detection algorithms have also been proposed to differentiate the fake from the real. To summarize the representative work for deepfake creation and detection, we have broadly categorized the algorithms into two categories in which single modality (Image/Video/Audio) or multi-modalities are exploited in the detection framework. Table 1 tabulates the key focus areas and limitations of the existing surveys to highlight the advantages and suitability of the proposed survey.

1.2. Motivation

The motivation of this review is to categorize the various deepfake detection algorithms based on their feature extraction techniques and key algorithms utilized in detecting fake outcomes. Based on the input data, deepfake algorithms are categorized either as single-modal techniques or multi-modal techniques. Based on the extracted features, single-modal techniques are categorized either as conventional methods or advanced methods. Conventional methods extracted handcrafted features and processed them using DL models. On the other hand, ML-based conventional methods classified the deepfakes using ML classifiers in the final stage. Complementary handcrafted features were fused using either score level or feature level fusion to generate unified robust features for deepfake classification. Furthermore, advanced methods categorized algorithms either as DL-based or hybrid. DL-based algorithms utilized DL approaches both at the feature extraction and processing stage. On the other hand, hybrid algorithms extracted features using DL approaches, fused and classified them using either ensembled ML or DL classifier. Nevertheless, multi-modal techniques are also categorized either as DL-based algorithms or ensembled algorithms based on the exploited feature extraction and processing methodologies. Ensembled algorithms fused the outcome from different DL models to generate robust features that can be fed to the classifier for deepfake classification. We believe that this is the first review of its kind which critically analyzed the representative work in the deepfake detection domain from different perspectives of exploited feature extraction and processing techniques.

1.3. Significant contributions

The key contributions of the paper are as follows:

- Firstly, the deepfake detection framework is broadly categorized into two classes as single-modal and multimodal detection approaches. Single-modal approaches are further categorized as conventional and advanced techniques whereas multimodal approaches into deep learning and ensembled fusion-based techniques to understand their salient features, similarities, differences, and applications.
- Secondly, statistical analysis has been conducted to explore the current trends in single/multi-modal deepfake detection algorithms. Single-modal and multi-modal are further categorized to review the shifting trends in traditional, ensembled, and multimodal-based fusion methodologies.
- Thirdly, deepfake creation techniques are reviewed following the multimedia content namely, audio, images, and videos. The techniques are thoroughly investigated to understand the process of creation of realistic content in order to explore robust deepfake detection algorithms.
- Lastly, the recommendations, suggestions, and limitations are thoroughly examined to highlight the need for generalizability and explainability of advanced deepfake detection algorithms to deal with practical challenges involved in its detection.

Table 1
Comparative analysis of the proposed survey with existing survey articles considering their key strengths and limitations.

SN	Survey papers	Strength	Limitations	Comparison with the proposed survey
1.	Albahar and Almaki [56]	Discussed various deepfake detection methods such as face detection, multimedia forensics, watermarking and CNN.	Limited historical perspective and narrow discussion on the key benefits of deepfake detection methods.	Statistical exploration of historical deepfake detection methods and categorization into multiple classes to exhibit clarity and identify potential research directions.
2.	Seow et al. [2]	Comprehensive overview of deepfake detection and generation methods for facial images and videos.	Limited focus on deepfake detection challenges in practical scenarios including audio in videos.	Emphasize the robustness and reliability of deepfake detection algorithms to address societal, ethical, and privacy concerns.
3.	Rana et al. [54]	Systematic review of deepfake detection videos under various categories.	Lacked in discussion for the deepfake detection models generalizability, explainability and ethical implications.	Detail the requirements of the deepfake detection model's generalizability and explainability to address the real-time challenges in deepfake detection algorithms.
4.	Malik et al. [57]	Categorized deepfake creation methods and discussed DNN-based deepfake detection methods.	Discussed DNN methods for deepfake detection in facial images/videos only.	Elaborate deepfake detection in various multimedia contents namely, audio, video and images with a wide range of methodologies namely, conventional, ML, DL and fusion-based approaches.
5.	Gupta et al. [58]	Reviewed deepfake detection algorithms utilizing ML and modality fusion techniques.	Analyzed deepfake video and images for detection, with limited discussion on deepfake creation.	Review a wider perspective of multimedia content including audio with images/videos in detection and creation methodologies.
6.	Mary and Edison [59]	Focused on DL-based models for detecting deepfake videos/images	Discussed very limited DL-based approaches exploiting CNN, RNN and LSTM.	Provide a wider perspective of deepfakes in DL methods including advanced approaches such as ViT, swim transformer, etc.
7.	Patel et al. [1]	Discussed deepfake media content generation and detection methods based on ML, DL ensembles, and multimodal techniques.	Limitedly discussed the handcrafted features-based methods for deepfake detection.	Summarize the wider perspective of single-modal and multi-modal deepfake detection algorithms which are further categorized for better understanding and clarity.
8.	Passos et al. [55]	Discussed DL-based deepfake detection methods in spatial and temporal domains under various categories for videos.	Restricted to DL-based detection methodologies for deepfake videos with brief discussion for deepfake audio detection.	Exhaustive discussion on deepfake detection algorithms for various media contents using conventional, ML, DL, ensembled, multimodal, fusion-based techniques for covering wider applications.
9.	Abbas and Taeihagh [20]	Reviewed and categorized the deepfakes generation and detection techniques using ML, and DL with a special focus on GANs.	Discussion on multi-modal approaches for realistic scenarios was not presented.	Categorize single modal and multimodal deepfakes detection. Reviewed multimodal detection techniques as DL and ensembled fusion-based approaches.

[ML: Machine learning, DL: Deep learning, CNN: Convolutional neural network, RNN: Recurrent neural network, LSTM: Long short-term memory, GANs: Generative adversarial networks, ViT: Vision Transformer].

1.4. Paper organization

The rest of the paper is divided into eight sections. Section 2 details the inclusion and exclusion strategy for selecting papers for this survey along with their statistical analysis. Deepfakes creation and detection methodologies are elaborated and categorized in Section 3. The details about the conventional and advanced techniques for deepfake detection are summarized in Sections 4 and 5, respectively. Section 6 reviews the existing work in the multi-modal deepfake detection category. The benefits, limitations, recommendations, and benchmarking of the proposed review are presented in Section 7. Lastly, the concluding remarks and future directions are provided in Section 8. For better clarity, the details about the organization of the paper are further illustrated in Fig. 1.

2. Search strategy and statistical distribution of deepfake algorithms

In this section, we have discussed the adopted search strategy for selecting the representative work for this literature review. We have also analyzed the statistical distribution of the selected publications to provide clarity of existing progress in the domain. The next section discusses the article selection strategy utilized in this work.

2.1. PRISMA strategy for article selection

A detailed search was conducted to select the most appropriate articles using various online platforms including IEEE, Sciencedirect, GoogleScholar, Springer, and other sources (MDPI, ArXiv, ResearchGate, etc.). The keywords utilized for exploring the articles were "DL-based deepfake algorithms", "AI-based deepfake algorithms", "ML-based deepfake algorithms", "multi-modal deepfake detection approaches", "Handcrafted features based deepfake detection algorithms", "transformer-based deepfake detection algorithms", "hybrid multi-modal deepfake detection algorithms", and "ensembled deepfake detection algorithms". In addition, good impact factor journals and reputed conferences from the year 2019 to till date are searched for finding relevant and good articles written in English. Fig. 2 illustrates the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) strategy along with adopted selection criteria to select critical studies for this work. Based on the above-mentioned criteria an exhaustive search was performed and a total of 6658 studies were identified. Duplicate articles were identified and removed by exploiting the "Find Duplicates" feature of EndNote software by Clarivate Analytics. After this step, a total of 2860 studies were retained. Three exclusion criteria were selected as (i) E1: Non-significant articles (ii) E2: Non-relevant articles (iii) E3: Articles with insufficient data. E1 excludes articles that did not utilize the considered methodologies and techniques utilized in this review. E2 removed those that were not related to the domain and E3 excluded those that did not have sufficient information in the discussion, processing data, and adopted methodologies. In addition, the articles from non-quality journals and conferences were also excluded from the mentioned exclusion criterion. The criteria that were considered for deciding the conference quality such as ranking, organizing institution, location, and previous records.

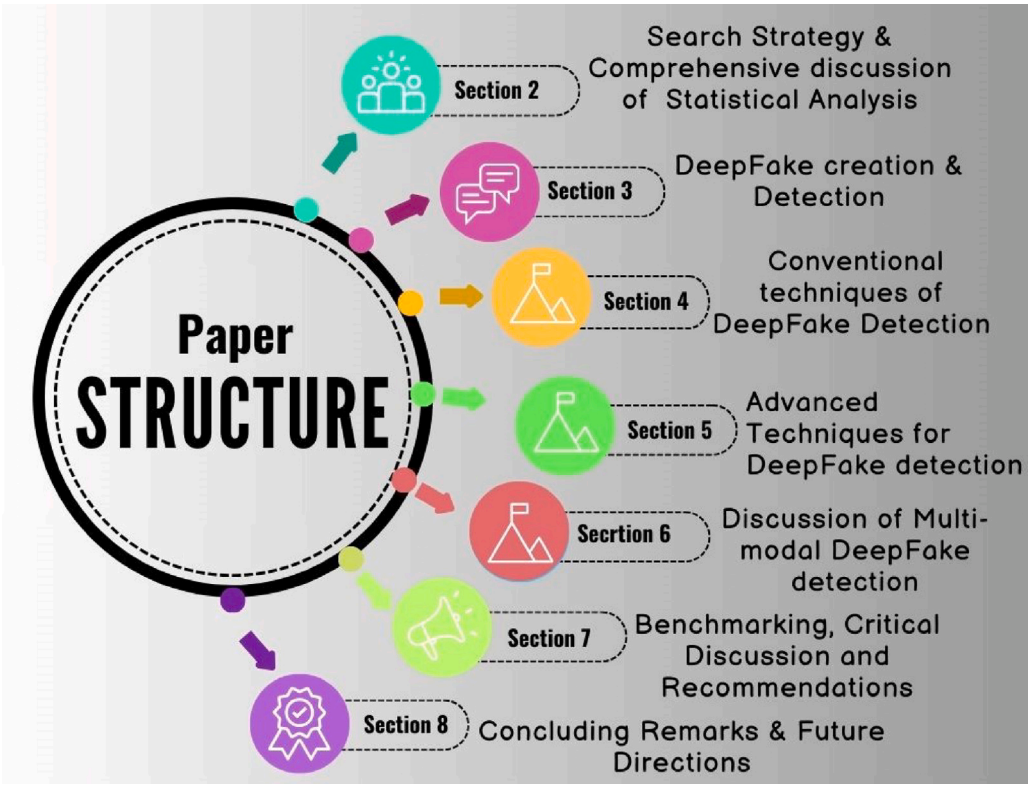


Fig. 1. Description of paper structure.

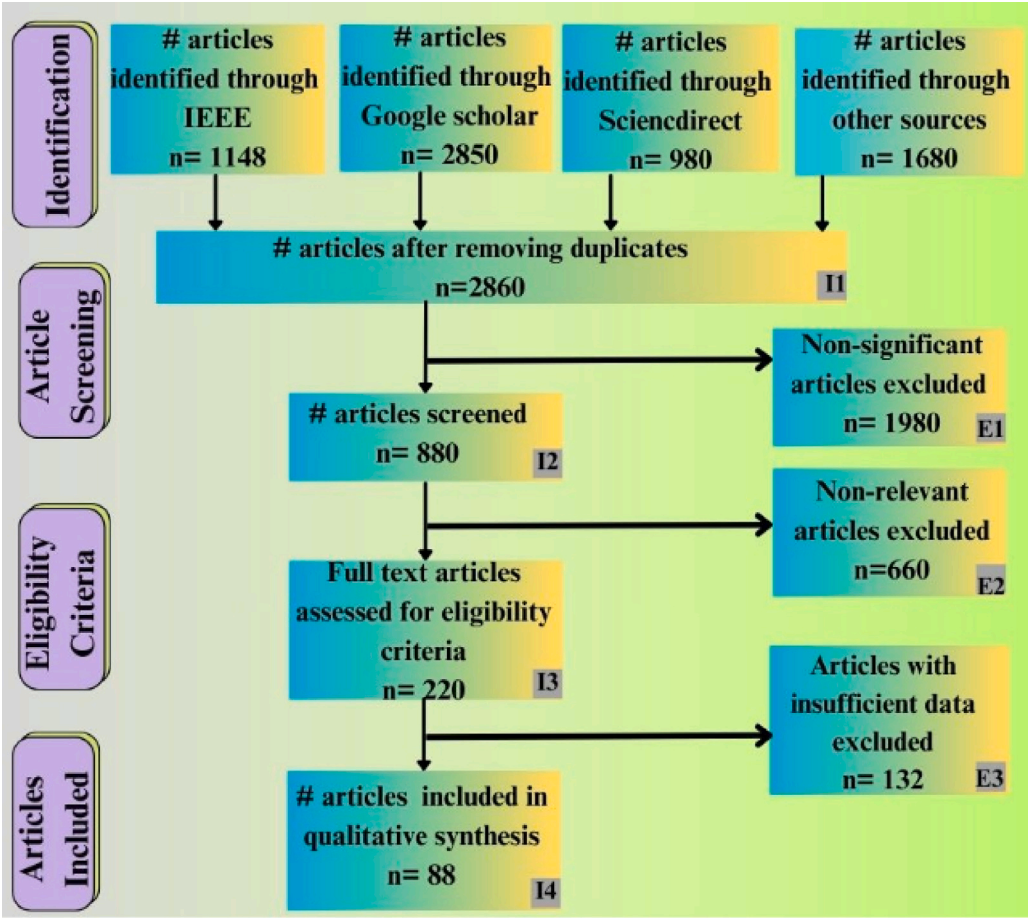


Fig. 2. PRIMA search strategy for study selection.

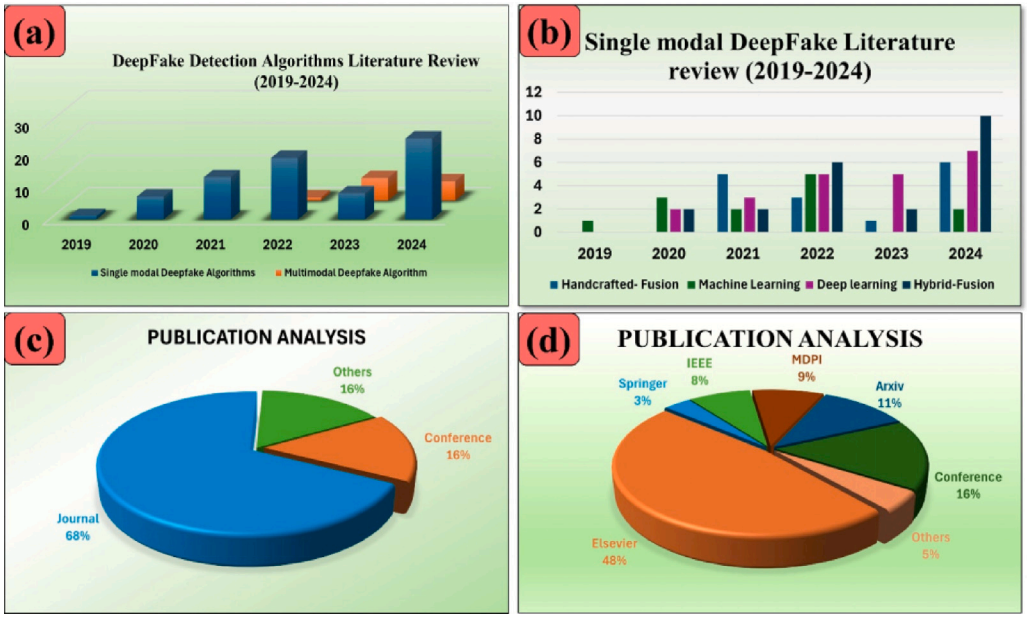


Fig. 3. Statistical distribution of deepfake detection algorithms: (a) Literature review of selected studies from year 2019 to 2024, (b) Literature review analysis of single modal deepfake literature from year 2019 to 2024, (c) Publication analysis as per the publication type, and (d) Publication analysis as per publisher.

Survey papers were also excluded from the literature review. Considering all these conditions in all three steps, the number of articles selected for exclusion from the analysis was 1980, 660, and 132, respectively. Finally, a total of 88 most relevant and critical studies were selected for discussion in this study. The next section will detail the statistical distribution of the selected studies from different perspectives.

2.2. Statistical distribution analysis of selected articles

The study focussed on the analysis of existing work published for deepfake detection either as single-modal deepfake algorithms or multi-modal deepfake algorithms. It is necessary to investigate the progress in the domain from past to present to understand future trends. This assessment will help to determine the contribution of these algorithms to the field of deepfake detection algorithms.

Fig. 3 depicts the statistical distribution of deepfake algorithms from the year 2019 to 2024. Fig. 3(a) illustrates that very few single deepfake algorithms are published from 2019 to 2021. These algorithms regain popularity in the year 2024 again. Publication trends evident that multi-modal detection evolved in the past few years only. Multi-modal algorithms are published from 2022 to 2024. The publication trends in single modal detection algorithms are further analyzed and depicted in Fig. 3(b). The analysis infers that there is a steep rise in several publications under DL-based and hybrid models from the year 2019 to 2024. Publication trends show that a comparable number of handcrafted features based on deepfake detection are also proposed due to computationally easy feature extraction and processing. Fig. 3(c) analysis illustrates that the contribution of journal publication (68%) is quite high in comparison to conference publication (16%) and others (16%). Further, the contribution of journal publication is divided as Elsevier (48%) [2,6,25,31,36,55,60,61], Springer (3%), IEEE (8%), MDPI (9%) [52,62,63], Arxiv (11%) [29,53,64,65], and others (5%) [33,66,67] as shown in Fig. 3(d). The highest number of journal articles are published by Elsevier publishers and others include the open access journals [24,68] and publications published by university journals [26,33,69,70] and CVPR conferences [71,72].

Statistical distribution of substantial parameters such as journal quality, number of datasets, model architectural parameters, and performance metrics from selected deepfake studies are illustrated in Fig. 4. Fig. 4(a) illustrates that most of the studies (41%) are published by various journals and conferences with low impact factor [22,28–30,51,53,73,74]. The journals with an impact factor from 2 to 4 contribute to 26% [24,52,62,63,67,68,75]. Studies (7%) are from journals above 8 impact factor [39,50,76–78]. The analysis of exploited datasets by various studies is depicted in Fig. 4(b). Most of the studies (50%) utilized up to three datasets [10,38,43,72,75,77,79–82]. Only 2% of the studies exploited 5 or above 5 datasets for experimental evaluation of the model [33,52,61,71,78,83,84]. Fig. 4(c) analyses the considered model architectural parameters in the selected studies. Very few studies (3%) have discussed the model explainability in their analysis [6,29,75,85–87]. Model generalizability by experimenting with its performance on cross-datasets is evaluated by studies (19%) [21,24,33,40,51,53,65,77,82,88–91]. Only studies (11%) [25,33,47,61,65,72,77,91–94] have considered data augmentation techniques either for improving the dataset quality or expanding its training size. After this, the considered performance metrics parameters are statistically analyzed in Fig. 4(d). Most of the studies utilized accuracy (32%) [7,25,26,40,66,68,71,72,77,81,82,86,88,89,92,95–98] and area under the curve (23%) [3,9,24,32,40,45,47,60–62,68,70,72,77,91,98] for comparing model performance with other existing methods. Very few studies considered sensitivity (2%) [8,26,62,86,95] and specificity (2%) [8,26,62,63,86] as performance parameters in the analysis. Also, confusion matrix (6%) [8,9,24,25,39,40,61–63,68,85,89,95,98] and F1-score (12%) [8,22,24,26,36,38,49,50,62,67,68,70,74,85,86,88,97–100] are exploited by limited studies in their experimental evaluation. The next section will detail the deepfake creation and detection techniques.

3. Deepfakes creation and detection

In this section, deepfake generation and detection methods for various multimedia content namely, image, video, and audio are discussed. DL-based methods have a wide range of applications in creating high-quality, professional, and tough-to-detect deepfakes. The details of deepfake creation methods are as follows.

3.1. DeepFake creation algorithms

Fig. 5 illustrates the various deepfake creation methods in accordance with various multimedia contents. For generating deepfakes images, DL-based networks such as GANs [101] and improved GANs [102,

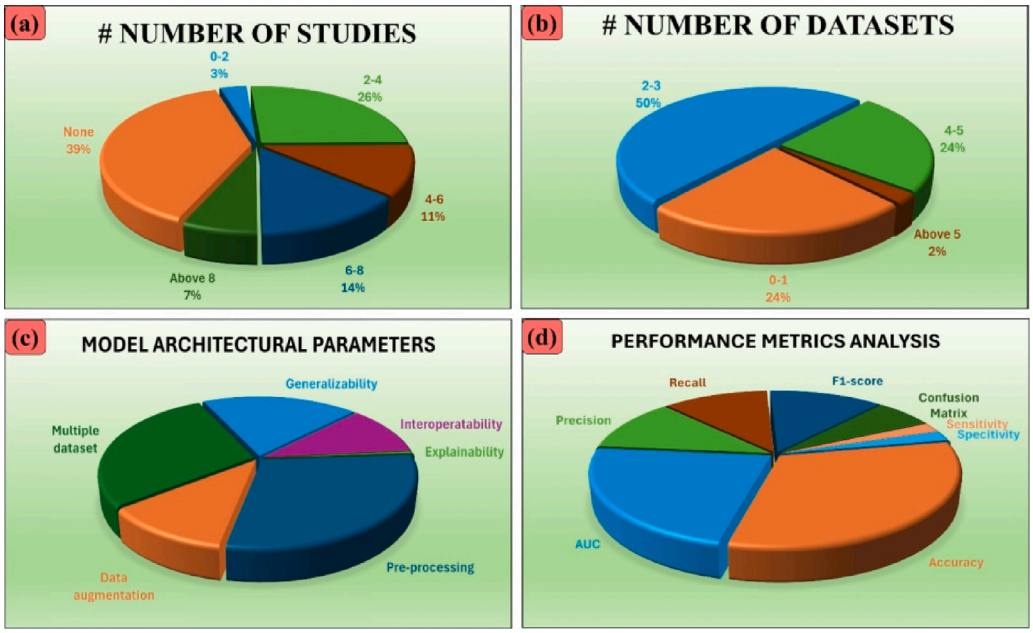


Fig. 4. Statistical distribution of selected studies (a) Analysis based on journal's impact factor, (b) Analysis based on the number of datasets utilized for performance comparison, (c) Analysis based on model architectural parameters utilized by selected studies, (d) analysis based on performance metrics utilized for performance comparison in the studies.

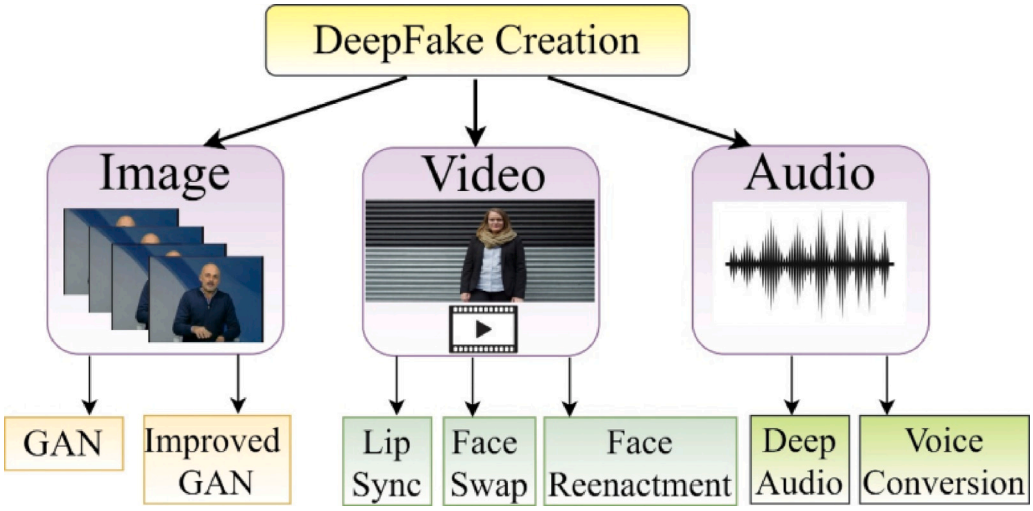


Fig. 5. Different techniques for creating DeepFake contents [GAN: Generative adversarial network, MFCC: Mel-Frequency Cepstral Coefficient].

103] are exploited. Lip sync, face swap, and motion reenactment methodologies are utilized for generating deepfake videos [55,104, 105]. Deepfake audios are created using deep audio techniques, voice conversion, and impersonation [106–108].

In [101], authors utilized super-resolution GAN for generating photo-realistic fake images. The model adopted ResNet with skip connection and perceptual loss function. This loss was generated from high-level feature maps of the VGG network invariant to change in pixel space. On the other hand, improved GANs namely AttGAN [102] and StyleGAN [103] were exploited by authors for generating high-quality and realistic photo images. AttGAN focussed on manipulating selected single or multiple facial points while keeping the other attributes the same. This network employed attribute classification constraint, reconstruction learning, and adversarial learning for high-quality editing of facial attributes. Facial attributes were edited by analyzing the relationship between the selected attribute and the latent representation of the face. On the other hand, StyleGAN studied variations in latent factors for generating high-quality fake images. High-level style features and stochastic effects were analyzed by StyleGAN in comparison to conventional GAN architecture for generating realistic fake images. Also, increasing the network depth not only improves image quality but also prevents the severity of separability loss.

The most commonly utilized techniques for generating deepfake videos are face swapping [105,109], lip-syncing [104,110], and face reenactment [111,112]. In [105], authors performed face swapping by editing latent spaces for facial and hair regions independently. Visual attributes that were to be faked, identified, and replaced within the latent space representation of the face and hairs. The entire image was reconstructed automatically after performing the face swapping. Authors introduced DeepFaceLab for face swapping and generating high-quality deepfakes [109]. DeepFaceLab is an open-source, and flexible code that performs face detection and then determines facial landmarks for performing the alignment. After this segmentation was performed which was followed by a conversion step to perform the face swapping. The method generated cinema-quality results and high-quality forgery video clips. Lip-synching can be used for generating

dubbed voice videos, virtual assistance, and video conferencing. KR et al. [104] generated deepfake videos using speech-to-speech and face-to-face translations in a visual module known as LipGAN. This module generated realistic talking faces videos by performing lip synchronization translating talking faces into the target language automatically. Authors performed lip sync by representing audio, modeling faces, and generating natural talking faces [110]. Audio–visual synchronization was analyzed using SyncNet to produce realistic and high-quality talking face videos. Another methodology for producing deepfake videos is face reenactment in which monocular videos were rendered with target videos for producing real-time deepfakes [111]. For performing reenactment, facial expressions were captured for both source and target using dense photometric consistency measure. Then, the target video was blended with the source for synthesizing the deepfake video stream. Zhang et al. [112] presented FReeNet for performing face reenactment by transferring the facial expressions from selected reference videos to the target videos. The framework consisted of an encoder–decoder in a unified landmark converter and then finally, the reference was converted to a target using a geometry-aware generator. The model learned the appearance and geometry of the target from the source video for generating high-quality and realistic deepfake videos.

Deepfake audios are created by superimposing reference audio on the target audio. Deep audio networks such as Char2Wav, Deep Vocie3, MelNet, and many others were analyzed for synthesizing fake audio [106]. These frameworks consisted of three modules namely, encoder, decoder, and converter. The encoder encoded text into key and value vectors as per timestep information. The decoder predicted the time-aligned mel-spectrogram from the key and value vectors of the audio samples using L1 and binary cross entropy loss. The converter stage computed the required parameters from the decoder for generating the deepfake audio. In [107], authors discussed speech synthesis for voice conversion and impersonation. Artificial audio can be generated digitally by concatenating the speech samples of the reference speaker with the target speaker. This reassembling of voice signals produced high-quality deepfake audio. Waveform synthesis-based speech synthesis systems create natural and realistic deepfake audio in comparison to feature synthesis.

To summarize, DL-based methods can create deepfakes from various multimedia content. These contents are generated for addressing the individual or system problems by manipulating the data in such a way that cannot be detected easily. With the advancement in technology, the negative and unauthorized use of deepfake content makes its detection essential. It is challenging to classify the deepfakes from the real ones as the generated content resembles similar to the original content. Hence, deepfake detection is essential to protect the society from its adverse impact. The details about the various deepfake detection algorithms are as follows.

3.2. DeepFake detection algorithms

In this subsection, we have categorized the various deepfake detection methods either as single-modal or multi-modal. Single-modal detection methods are further classified as conventional techniques or advanced techniques. Based on the feature extraction methods and processing algorithms, conventional techniques are explored either as handcrafted feature-based algorithms [42–44,72,79,87] or ML-based algorithms [10,21,22,28,30,33,73,94]. On the other advanced techniques are categorized as DL-based algorithms [6,34–36,66,84,113,114] and ensembled algorithms [37,38,67,68,80,115–117]. Further, the multimodal detection methods are analyzed either as DL-based techniques [50,64,82,83,118] or hybrid techniques [39,49,51,93]. Fig. 6 illustrates the categorization of deepfake detection algorithms reviewed, analyzed, and detailed in this study.

The key difference between the single-modal and multi-modal deepfake detection techniques is the number of input modalities exploited in the detection model [39]. Single-modal deepfake detection algorithms consider individual modalities namely, image, audio and video for feature extraction to detect fake or manipulated content. On the other hand, multi-modal deepfake detection algorithms exploited the synchronization between audio and video, lip movement, facial movements, mouth shape dynamics, and spoken phenomics to detect false contents altogether. However, the deepfake creation techniques address this change efficiently by generating realistic videos in which it is hard to detect the difference between real and fake. Single modalities are not efficient enough to detect these inconsistencies whereas multi modalities deepfake detection models can analyze the variations in speech and facial characteristics effectively.

In comparison to multi-modal deepfake detection techniques, single-modal deepfake detection algorithms have a wide range of applications. The reason could be the availability of public datasets, less computational complexity, and generalizable models. However single modality-based models are easily deceivable. Multimodal deepfake detection algorithms can address the real-time challenges of deepfake detection efficiently, and are more accurate and robust. However, limited availability and low-quality content in audio–visual multimodal deepfake datasets restrict their practical applicability [83]. Also, the existing multimodal deepfake datasets labeled the entire audio–visual pairs as fake which makes it tedious to detect the real part from it [119]. These algorithms can perform emotional analysis, speech recognition, and facial analysis to compute the variations that are often present in deepfakes. The similarity between the perceived emotions between the two modalities i.e. audio and video in a video can be analyzed using multi-modal learning to detect deepfakes [120].

Further to highlight the strengths and weaknesses of deepfake detection algorithms under each category, the salient features are elaborated in Table 2. The various deepfake detection algorithms are compared based on the exploited methodology, feature extraction, data requirements, data preprocessing, and many other substantial parameters. The similarities and differences of each category of deepfake detection algorithms are tabulated in Table 3. In this, the algorithms are compared based on the utilized AI models, fusion strategy, data classification stage, model explainability, cross-datasets evaluation for model generalizability, and many others.

4. Single modal based conventional techniques for deepfake detection

Fig. 7 illustrates the generalized framework of conventional techniques for deepfake detection. The preprocessing stage extracted frames from video using frame sequencing, and audio data was converted to an image by generating a spectrogram for extracting robust features for the processing stage. Based on the extracted features and processing stage conventional techniques are categorized as handcrafted features-based algorithms and ML-based algorithms. The details about the handcrafted features-based deepfake algorithms are discussed in the next section.

4.1. Handcrafted features-based fusion algorithms

Table 4 tabulates the salient features of representative work selected under the category of handcrafted feature-based approach. These approaches performed manual feature extraction to maintain the computational complexity of the model. The extracted features can be color, texture, HOG, computer vision (CV), etc.

Handcrafted features are lightweight, less complex, and easier to extract. In this direction, authors extracted texture features to analyze the irregularities in the different regions of the deepfakes [43]. The texture feature was extracted using the LBP (local binary pattern) approach. The feature was extracted for the facial region only which was detected using the MTCNN classifier. Gamma correction and inter-frame face similarity were also used for detecting facial regions in low-contract images. Similarly, Yang et al. [121] studied the difference between the

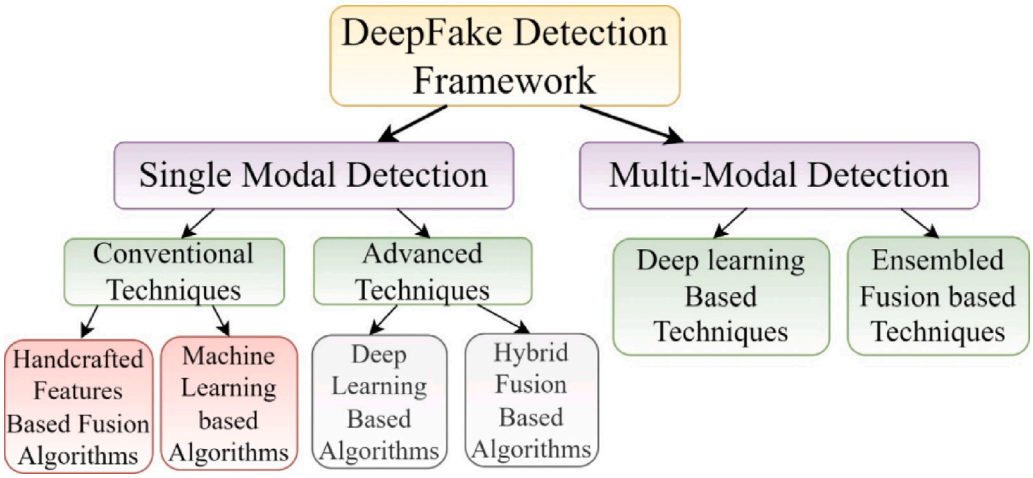


Fig. 6. Categorization of various deepfake detection algorithms.

Table 2
Salient features of various deepfake detection algorithms.

SNo.	Attributes	SMD algorithms		MMD algorithms	
1	Methodology	CT	AT	DL^T	HT
2	Feature extraction	HC or DF	HL	DF	DF or HIE
3	Data requirements	Less	High	High	High
4	Data preprocessing	✓	×	×	×
5	Performance	Moderate	High	High	High
6	Advantages	Require less training data	Automatic feature extraction	Identified visual inconsistencies	Suitable for real-world applications
7	Limitations	Not suitable for real-time	High training data	High training data	Slow processing
8	Computational complexity	Less	High	Moderate	High

SMD: Single modal detection, MMD: Multi modal detection, CT: Conventional technique, AT: Advanced technique, DL^T: DL technique, HT: Hybrid technique, HC: Handcrafted, DF: Deep features, HL: High-level, HIE: Hierarchical.

Table 3
Similarities/Differences between various deepfake detection techniques.

SNo.	Attributes	CT	AT	DL^T	HT
1	AI-models	ML	DL and Transformer	DL	DL and Transformer
2	Fusion strategy	✓	✓	×	✓
3	Data classification	ML-based	DL-based	DL-based	ML or DL based
4	Multiple datasets	✓	✓	✓	✓
5	Statistical measures	✓	✓	✓	✓
6	Explainability	×	✓	✓	✓
7	Cross-dataset evaluation	✓	✓	✓	×
8	Interoperability	×	✓	×	✓

DL: Deep learning, ML: Machine learning, CT: Conventional technique, AT: Advanced technique, DL^T: Deep Learning technique, HT: Hybrid technique.

texture of real and fake images in an image saliency map. Superpixel segmentation was utilized for computing the texture information. The Euclidean distance was computed between the corresponding regions using three different approaches RGB, CIELab, and LBP. The image was divided into multiple superpixels for creating the saliency. On the other hand, authors ensembled texture and artifact information for detecting the inconsistencies between the real and deepfakes [42]. Artifact refers to the random and synthesized noise disturbances present in the real and deepfakes. The cosine distance measured the similarity distance between the texture inconsistencies between real and fake. Score level fusion fused the texture inconsistencies score and artifacts to obtain the combined results for deepfake detection. Authors fused fisherface with LBP histogram features on the dimensionality reduced facial images for predicting deepfakes [26]. The noise in the images was removed using the Kalman filter and images were normalized for image contrast enhancement.

Analysis of illumination variations is highly effective for detecting deepfakes. For this, Zhu et al. [84] extracted illumination inconsistency at the frame level along with multi-level feature enhancement for enhancing feature-level illumination. Features in the low-frequency domain were obtained by DCT (Discrete cosine transform) for capturing illumination changes in the frequency domain. Feature level fusion fused to obtain the final score for deepfake detection. In [79], the author computed DCT values by dividing the detected face into a uniform block of 8 × 8. Once the face was localized then partly sign-altered value and partly sign-altered mean value was computed. Finally, deepfakes were detected using mixed modulation techniques. However, the authors extracted spatial and frequency domain features and fused them within a knowledge distillation for classifying the deepfakes and real [78]. The features were extracted by applying DCT on raw as well as compressed images. The extracted features were fused in three phases namely, split, fuse, and select using SKAttention. Xu et al. [47] proposed joint distillation along with the knowledge transfer stage for deepfake detection. The feature importance was determined by exploring the relationship between gradient information and feature maps. The negative transfer in feature importance was adjusted using the exponential decay rule. Authors employed knowledge distillation along with representation learning for classifying deepfakes [72]. Features were represented using transfer learning to reduce the computational complexity of the base model. Adaptation learning was utilized for

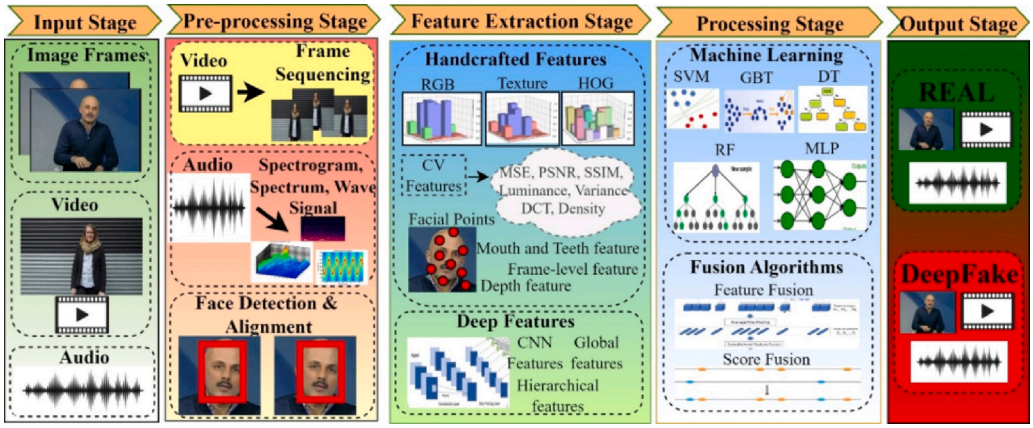


Fig. 7. Generalized architecture of conventional techniques for deepfake detection.

controlling the three loss functions namely, KD loss, cross-entropy loss, and feature-based square loss used for calculating the difference between real and fake.

To determine the integrity of videos, authors proposed to extract the CV features from each video frame [81]. The rate of change between frames was computed for each feature to distinguish the deepfakes from the real. Similarly, Saravana et al. [46] also extracted multiple CV features but using fuzzy clustering algorithms. Images were pre-processed using the Gabor filter-based Gaussian rule to improve quality by eliminating noises. Deep belief network-based classifier for classifying the deepfakes from real by reducing the detection loss and improving the energy function. However, the authors extracted semantic and contextual information to detect deepfakes generated by the face-swapping technique [77]. Semantic information included the boundaries of facial regions whereas contextual features captured the face contexts such as hair, ear, etc.

Biological and physiological features are considered to be the best visual descriptors for detecting deepfakes. In this direction, Elhassan et al. [45] utilized teeth and mouth movement for detecting the deepfake videos. 68 landmarks on facial images with open mouth showing teeth acted as distinguished and lightweight features were identified for detecting the fake teeth. Mouth regions were cropped from facial images using mouth aspect ratio and mouth openness with 12 points for both upper and lower lips. Features were processed using multiple DL algorithms and deep fakes were classified using clustering algorithms. On the other hand, authors analyzed the change in skin color as a physiological feature in deepfake images using remote photoplethysmography (rPPG) through camera [85]. rPPG were computed for the extracted region using the Chrom method which averages the RGB values. These features were converted to the frequency domain using the Fast Fourier transform and frequency composition was identified using power spectral density. This method effectively classified the deepfakes from the real.

Spoofed audio and speech need to be determined to prevent deception and voice phishing. In order to determine spoofed audio, authors fused raw waveform signals with spectrograms as distinguished features [87]. The feature distribution of spoofed speech for explaining its likelihood was analyzed using the SHAP method of model explainability. Feature correlation in a high-frequency range was used for the classification of real and deepfake audio. On the other hand, Ballesteros et al. [92] extracted entropy-based handcrafted features from the audio recordings. These features were transformed into histograms for classifying deepfake audios from the real.

To summarize, handcrafted features were extracted and processed for identifying deepfakes audio, video, and images. The differences and the variations between the real and the spoofed information were identified and compared for classification. Data was pre-processed not only to identify the region of interest but also to eliminate the impact of noise and improve contrast and resolution. The extracted information was processed and classified using either DL models or ML models. Handcrafted features based on conventional techniques are fast and suitable for real-time but with limited applicability.

4.2. Machine learning-based algorithms

In this subsection, ML-based deepfake detection techniques are detailed. The techniques are categorized as ML-based deepfake detection in which features are extracted using either a handcrafted approach or DL-based approaches but classified using various ML-based techniques. Table 5 tabulates the salient features of ML-based approaches for deepfake detection. ML-based approaches for deepfake detection can be trained easily and faster in providing the results. In this direction, Wang et al. [31] trained an RF classifier for detecting the deepfakes. Deepfakes have fewer facial point features in comparison to real ones which were efficiently classified with ML based classifier from a fused facial region descriptor. The facial region descriptor was generated by integrating various handcrafted features such as SIFT (Scale-invariant feature transform), SURF, FAST (Features from accelerated segment test) & BRIEF (Binary robust independent elementary features), ORB (Rotated BRIEF) and A-KAZE. RF also selected important features to improve the efficiency of the detection algorithm. On the other hand, the SVM classifier was utilized for detecting variations in facial regions considering inconsistencies of texture features [10] and head poses [32]. In [10], authors trained and classified the deepfake detection model using SVM. Texture feature maps were generated for both real and fake videos at three levels namely, pixel, block, and region. The statistical disparities were analyzed by extracting the color information from the R, G, B, V, and Y channels. However, the authors estimated the errors in 3D head poses of facial images for detecting deepfakes [32]. The facial landmark points were formulated as an optimization problem and solved using the Levenberg–Marquardt algorithm. SVM classifier was trained by estimating the difference between full facial landmarks and central facial region landmarks for classifying deepfakes from real.

Multiple ML algorithms were trained to obtain better results for deepfake detection. In this direction, Hamza et al. [28] exploited RF, MLP (Multi-layer perceptron), DT, EXC (Extra tree classifier), LR (Logistic regression), Gaussian NB (Naïve Bayes), AdaBoost, GB (Gradient boosting), XGBoost, LDA (linear discriminant analysis, and QDA (Quadratic discriminant analysis). Overall, SVM outperformed the other ML algorithms and achieved the highest accuracy. Similarly, authors adopted six ML algorithms such as SVM, MLP, DT, LR, NB, and XGBoost for classifying the deepfake audio from real [30]. The processing for these classifiers by selecting important features for training using PCA (Principal component analysis). This model also

Table 4
Single modal conventional fusion based deepfake detection techniques with handcrafted features.

SN	Citation	Features	Modality	Fusion strategy	Dataset	Performance metrics	Summary
1.	Hsu et al. [79]	DCT	Image	Modulation	CVG-UGR, Large Logo DB, and Large Logo DB	PSNR, SSIM, BER, Loss and NCC	• Identified relative positions of modification in real images using temper detection mean value. • Optimized network parameters using gray wolf optimizer. • Adopted denoising mechanism for extracting watermark information.
2.	Zhu et al. [84]	Frame-level illumination variations	Image	Attentional based feature fusion	FaceForensics++, Celeb-DF and DFDC	ACC and AUC	• Analyzed illumination inconsistencies in inter-frame for deepfake detection. • Multi-level feature enhancement for enhancing inconsistencies at feature-level. • Results of illumination decomposition not good in regions where hair were present.
3.	Leporoni et al. [44]	RGB and Depth	Image	Attentional based feature fusion	FaceForensics++	ACC, AUC, and Heatmaps	• Extracted depth information from whole image frame using pre-trained model. • Two parallel networks for depth features and one network for RGB extraction. • Both features processed individually and stacked together using MobileNet v2.
4.	Wang et al. [78]	Frequency domain and spatial-domain features	Image	Attentional based fusion	FaceForensics++, and Celeb-DF	ACC and AUC	• Fused the spatial and frequency domain features to improve detection ACC. • Knowledge distillation at the classification step transferring knowledge for classifying deepfakes. • Achieved highest ACC (98.98%) on Celeb-DF DB.
5.	Yu et al. [87]	Raw waveform signals and spectrograms in vector stitching	Audio	Feature fusion	ASVspoof 2019	Loss rate and SHAP value distribution	• Fused extracted features using lighter feature level fusion techniques. • Utilized explainability to determine feature contribution and importance. • Analyzed feature correlation in high frequency range of the waveform.
6.	Gao et al. [42]	Texture and artifact	Video	Score level fusion	FaceForensics++, Celeb-DF, WildDF and DFDC	ACC, AUC, and CM	• Texture artifacts detector to detect the inconsistencies between the real and fake. • Multiple deformable convolutional stacks to separate between target and source using the soft trainable mask. • Captured artifacts data using the shallow network.
7.	Kim et al. [72]	TL-based feature representation	Image	Representation learning and KD	FaceForensics++	ACC, F1-score	• Extracted features presented in both source and target dataset. • Experimented on a low-quality dataset to prove the model efficiency. • Performance evaluated on a single small dataset and achieved 86.97% (ACC).
8.	Kingra et al. [43]	Texture	Image	Combining LBP with LBPNet	Celeb-DF, DFDC, and DeeperForensic	ACC, AUC, PR, and RE	• Utilized local binary pattern for extracting texture features from facial regions. • Fully connected deep CNN as classification network for deepfake. • DB augmentation for generating real-time images for deepfake classification.

(*continued on next page*)

Table 4 (*continued*).

9.	Ni et al. [85]	Physiological	Video	FFT	DeepFakeTIMIT	ACC, F1-score, CM and AUC	• Analyzed change in skin color using remote photoplethysmography signals. • Extracted skin color information using chrom method. • Generated matrix visualization heatmap using time–frequency domain analysis of signals.
10.	Lee and Kim [81]	CV features: mse, psnr, ssim, rgb, hsv, histogram, luminance, variance, edge_density, dot	Video	Grouped the rate of change between frames	FaceForensics++ and DFDC	ACC	• Detected the rate of change in the subsequent frame incorporated by GANs for generating deepfakes. • Utilized multiple CV features for detection. • Achieved avg. 96.53% (ACC) on multiple datasets.
11.	Nirkin et al. [77]	Context and semantic	Image	Simple score concatenation	FaceForensics++, Celeb-DF-v2, DFDC and SG DB	ACC, AUC, FPR	• Compared changed context information in deepfake with real for detection. • Trained face discrepancy network to compare the identity information in face and its context. • Performed analysis on unseen data to determine model ACC.
12.	Yang et al. [121]	Texture	Image	Guided filter combining linear relationship	FaceForensics++	ACC	• Detected deepfake by comparing the texture difference between the real and fake image. • Guided filter to enhance texture information in the image saliency during pre-processing. • Achieved detection ACC (95%) on benchmark dataset.
13.	Ballesteros et al. [92]	Statistics, entropy and histogram features	Audio	CNN	LJ Speech DB, and SG	ACC, RE, PR and loss	• Transformed audio inputs into images such as spectrograms for classification. • CNN with data augmentation and dropout for fake audio classification.
14.	Saravana et al. [46]	CV features such as mse, PSNR, SSIM, rgb, hsv, histogram, luminance, variance, edge_density, DCT	Video	Contraceptive loss based pairwise learning	FaceForensics++ and DFDC	ACC, PR, RE, AUC, error and loss	• Extracted CV features using fuzzy C-means clustering. • Utilized stacked RBM layer in hidden layer along with sigmoid activation. • Adopted contrastive loss for handling outputs during pairwise learning. • Achieved global ACC (98.5%) across multiple DBs.
15.	Suganthi et al. [26]	FF-LBPH	Image	FF-LBPH-DBN	FFHQ, 100K-Faces, DFFD, and CASIA-WebFace	ACC, SPE, SEN, AUC, error rate and computational time	• Utilized DBN-RBM classifier for deepfake classification. • Dimensionality reduction algorithms for feature extraction. • Kalman filter for eliminating noise from the images.
16.	Xu et al. [47]	Gradient guided distillation feature map	Video	Xception_6S	FaceForensics++ and Celeb-DF	AUC, ACC, FLOPS, and Param	• Applied joint distillation loss function for transferring information from shallow to deep layers. • Dimension alignment for relating teacher network with the student network. • Analyzed model generalizability in cross-domain scenarios.

(*continued on next page*)

claimed that the SVM is superior and accurate in detecting real and fake video in a short time. Authors utilized multiple ML classifiers such as SVM, RF, extremely randomized tree, DT, MLP, Stochastic GB, and KNN for training the model for identifying deepfake videos [23]. Multiple handcrafted features were extracted and fused to form a single deepfake feature vector before feeding to the ML classifier. However, no feature-selected approach was adopted to improve the model processing.

In [71], authors extracted local features using expectation maximization algorithms. This algorithm generated a feature vector by computing probability and estimating weights for each element. This feature vector was integrated with RGB channels for obtaining the final vector for detecting deepfakes generated by multiple GANs networks. The deepfakes were classified using different ML algorithms

Table 4 (*continued*).

17.	Elhassan et al. [45]	Teeth and mouth features	Video	CNN, DenseNet121, DenseNet169, EfficientNetB0, EfficientNetB7, InceptionV3, MobileNet, MobileNetV2, ResNet50, vgg16, vgg19 and Xception	Vid-TIMIT, Celeb-DF, and UADFV	ACC, and AUC	• Model highly dependent on facial images with mouth open showing teeth. • Multiple DL models for determining the model suitability for processing features. • Utilized stochastic gradient for model optimization and faster convergence. • Among all the DL models, CNN achieved the highest ACC and AUC.

[DB: Dataset, ML: Machine learning, SG: Self-generated, KD: Knowledge distillation, FF: Fisher face, LBPH: Linear binary pattern histogram, ACC: Accuracy, PR: Precision, RE: Recall, AUC: Area under the curve, SPE: Specificity, SEN: Sensitivity, CV: Computer vision, GAN: Generative adversarial networks, mse: mean squared error, PSNR: peak signal to noise ratio, SSIM: structural similarity index measure, hsv: hue; saturation; value, DCT: Discrete cosine transform, FFT: Fast Fourier transform, BER: Bit error rate, NCC: normalized cross-correlation, DFDC: Deepfake detection challenge, DBN: Deep belief network, RBM: Restricted Boltzmann machine].

such as SVM, LDA, and KNN. Authors utilized three ML algorithms namely, SVM, RF, and DT along with five DL algorithms for detecting tempering in medical imaging [94]. Five DL models were used for feature extraction from the augmented datasets. Classifiers were trained to detect the tempered medical images using different ML classifiers. Among all the ML classifiers, SVM achieves the highest training and test accuracy. Farid et al. [33] proposed to utilize ML-based classifiers for detecting deepfakes in audio, video, and images. The synthetic content generated through face swap, lip sync, and other methodologies were easily classified from real using ML classifiers such as SVM and RF.

Further, authors utilized ML algorithms for detecting the deepfake images generated by face swapping algorithms [73]. The indicators such as skin details, color mismatch, temporal flickering, head positions, resolution, eye blinking rate, face warping artifacts, and lipsync were considered to detect deepfakes. For achieving better performance, noise was introduced using Gaussian blur, exponential blur, Rayleigh blur, and random specific blur. The video frames were analyzed, and the video was classified as deepfake if 1% of the frames were modified. Mitra et al. [22] exploited multiple CNNs for feature extraction and ML-classifiers for the identification of deepfake videos. The model computational efficiency was improved by extracting the key frames from each video to determine its manipulated contents. Similarly, authors utilized multiple CNNs for feature extraction and a classifier for detecting manipulated videos [21]. The frames were extracted from the videos, and faces were detected, cropped, and resized before processing. The features were extracted and processed through a global average pooling2D with fully connected followed by softmax as the classifier network. On the other hand, similarity scores were fed to the classifier network for identifying the video content as fake or authentic [8]. LR was used for converting the high similarity score into likelihood ratio for better calibration of analysis obtaining misleading evidence. The method was able to identify fake videos generated by face swap techniques for a portrayed person. However, the limited dataset availability for multiple reference videos of the same person restricts the cross-validation of the method.

To summarize, ML-based deepfake detection techniques are lightweight, interpretable, and understandable with less processing time. ML-based classifiers are easily trainable for detecting the deepfake contents generated using various creation techniques. Features were extracted using conventional techniques or DL-models for developing the robust model. The feature selection and keyframe selection approaches were adopted to improve the fused feature vector before feeding it to the ML-classifier.

5. Single modal-based advanced techniques for deepfake detection

In this section, advanced techniques for deepfake detection are elaborated using a single modality namely, image, audio, and videos. Advanced techniques utilized DL techniques for feature extraction, feature processing, and classification. Based on this approach, advanced techniques are categorized either as DL-based algorithms or hybrid algorithms. DL-based algorithms utilize DL approaches for feature extraction and feature classification whereas hybrid algorithms utilize multiple DL-approaches for feature extraction, processing, and classification. Multiple ML-algorithms are exploited for classification with DL-models for classification in hybrid algorithms. Fig. 8 illustrates the various components of deepfake detection using advanced techniques.

5.1. Deep learning-based algorithms

In this section, various DL-based algorithms for deepfake detection are discussed. Table 6 tabulates the representative work categorized under this category of deepfake detection algorithms. These algorithms are suitable for detecting deepfakes in highly compressed low-quality videos. In this direction, Zhu et al. [27] extracted spatial-frequency features using spatio-temporal inconsistency in highly compressed deepfake video. The temporal correlation information and structural knowledge were processed using the temporal-contrastive distillation model based on a two-stage student-teacher (ST) network. Similarly, authors utilized a two-branch convolutional auto encoder for performing deepfake reconstruction and classification [61] video. The temporal correlation information and structural knowledge were processed using the temporal-contrastive distillation model based on a two-stage student-teacher (ST) network. Similarly, authors utilized a two-branch convolutional auto encoder for performing deepfake reconstruction and classification [61]. Local and global information was combined and mapped the extracted features into latent space for improving the generalization ability. XceptionNet was exploited in the shared encoder branch and ReLu after each fully connected layer in the classifier network. Authors proposed convolutional ViT for deepfake video detection [65]. Convolutional Vit consisted of a CNN network for extracting learnable local and global features and ViT for processing these features. ViT utilized eight attentional mechanisms for categorizing the video either as real or deepfake. A softmax layer was applied to restrict the final weights between 0 to 1 for deepfake detection.

In [97], authors proposed a holistic DFD transformer for detecting deepfakes videos. Features namely, spatial, temporal, and spatiotemporal were extracted using 2D and 3D convolutional layers in the transformer, respectively. Self-attention-based transformer encoders processed the extracted features and fused the feature vectors followed by softmax layers. A fully connected linear classifier was used for classification using dropout and L2 regularization. Pu et al. [36] utilized a dual-level network for fusing the frame-level and video-level features. Spatial features were extracted using ResNet-50, pre-trained CNN. Temporal learning was performed using GRU nodes followed by average polling. After this, the frame level and video level classifier consisting of a fully connected layer classified the videos as deepfakes by minimizing the loss function. Similarly, authors extracted spatio-temporal features using 3D CNN by considering short sixteen consecutive frames from a

Table 5
Single modal conventional deepfake detection techniques with machine learning-based algorithms.

SN	Citation	Features	Modality	Technique	Dataset	Performance metrics	Summary
1.	Wang et al. [31]	Feature points from subdivided facial regions	Image	RF	DeepfakeTIMIT, UADFV, FF++_DeepFake DFD, DFDC, and Celeb-DF_V2	AUC and computational time	• Extracted multiple features from various facial points. • Calculated feature Gini importance for identifying crucial features for detection. • Training model RF quite faster in comparison to other techniques.
2.	Yang et al. [32]	Head poses	Image	SVM classifier	UADFV DB	AUC	• Detected face and 2D facial landmarks in the image. • Detected 68 facial landmarks using face detector DLib software package. • Difference between the head projections of fake and real determined by alignment error.
3.	Xia et al. [10]	Color and texture	Video	SVM classifier	FaceForensics++ and Celeb-DF	AUC, ACC and computational time	• Extracted RGB color feature and texture feature from co-occurrence matrix. • Designed three feature maps for capturing effective features. • Trained SVM classifier for binary classification of deepfake videos.
4.	Farid [33]	Low level (pixel-based) and high level (semantic features)	Image, audio, video	LR, SVM	ASVSpoof, FaceForensics++, Celeb-DF-v2, DFDC and deepfake DB	–	• Learning based detection algorithms classified real from fake content using ML classifiers. • Extracted artifacts-based low-level and high-level features fed into classifiers for detecting deepfakes.
5.	Hamza et al. [28]	MFCC	Audio	SVM, MLP, DT, ETC, LR, GNB, AB, GB, XGBoost, LDA, and QDA	for-rece, for-2-sec, for-norm and for-original	ACC	• Utilized ensembled feature approach, MFCC as the main feature. • Compared different ML algorithms for detecting deepfake from the real. • SVM outperformed other ML algorithms.
6.	Iqbal et al. [30]	MFCC, spectral_rolloff, spectral_centroid, spectral_contrast, spectral_bandwidth and zero_crossing_rate features	Audio	SVM, MLP, DT, LR, GNB, and XGBoost	Fake or real DB	ACC	• Utilized data framing for converting audio signals to machine-readable form. • Sliding window technique for extracting 270 features, and PCA technique to determine 65 important features. • XGBoost achieved average highest ACC (93%) compared to other ML models.
7.	Maksutov et al. [73]	–	Image	ML algorithms	Celeb-DF	AUC	• Analyzed deepfake quality to classify fake and real. • Analyzed color mismatch between fake and real for classification. • Applied affine transformation on randomly cropped and resized images.
8.	Rana et al. [23]	HOG, Haralic, Hu Moments, and CH	Video	SVM, RF, MLP, KNN, extremely randomized trees, DT, and Stochastic GB	FaceForensics++, Celeb-DF, VDFD and DFDC	ACC, AUC, PR, RE, F1-score, *p*-value, F-test and Q-test	• Adopted feature selection techniques to reduce the model complexity. • Extracted multiple features and fused them into a single vector. • Utilized various ML classifiers for detecting deepfakes.
9.	Mitra et al. [22]	Deep features	Video	ML classifier	FaceForensics++ and DFDC	ACC, PR, RE, and F1-score	• Compressed videos to reduce the model complexity. • Multiple DL models for extraction of robust features. • Simple classifier for classifying videos either as fake or real.

(*continued on next page*)

Table 5 (*continued*).

10.	Solaiyappan and Wen [94]	Deep features	Image	SVM, RF, and DT	CT DB, LIDC-IDRI and CT-GAN DB	ACC, AUC, PR, RE and Loss	• Multiple DL models for feature extraction and multiple ML classifiers for classifying fake and real medical images. • Utilized tempered and untampered DB to predict real and artificially modified medical images. • SVM achieved highest classification ACC in comparison to other models.
11.	Mitra et al. [21]	Deep features	Video	ML classifier	FaceForensics++	ACC	• Exploited TL for reducing time and computational complexity. • Replaced last layer for connected end-to-end fine-tuning. • Xception net-based features along with classifier achieved highest ACC (96%).
12.	Reis and Ribeiro [8]	Deep CNN based feature extraction	Video	Threshold based classifier	Celeb-DF (v2) and world leader DB	ACC, AUC, PR, RE and F1-score	• Extracted similarity scores from real and fake videos. • Threshold-based classification for classifying videos as fake or real. • Applied LR to convert scores to likelihood for forensic analysis.
13.	Guarnera et al. [71]	RGB and Local features using EM	Image	SVM, LDA and KNN	CELEBA DB and five different GANs generated DB	ACC	• Explored different kernel dimensions for detecting deepfakes. • Used multiple GANs for generating deepfake images. • Adhoc validation using different ML classifiers.

DB: Dataset, SVM: Support vector machine, ML: Machine learning, MFCC: Mel-frequency cepstral coefficients, TL: Transfer learning, LR: Logistic regression, MLP: Multi-layer perceptron, DT: Decision tree, ETC: Extra tree classifier, GNB: Gaussian Naive Bayes, KNN: k-Nearest neighbor, PCA: Principle component analysis, AB: Ada Boost, GB: Gradient boosting, LDA: Linear discriminant analysis, HOG: Histogram of gradient, CH: Color histogram, QDA: Quadratic discriminant analysis, DFDC: Deepfake detection challenge, ASVSpoof: Automatic Speaker Verification Spoofing and Countermeasures challenge, AUC: Area under the curve.

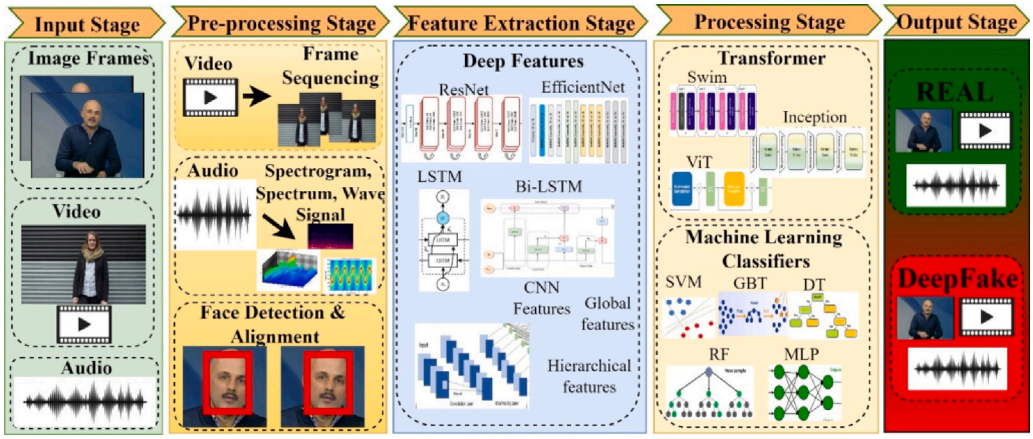


Fig. 8. Generalized architecture for deepfake detection using advanced techniques.

video [35]. The extracted features were processed using CNN consisting of a convolutional layer, dropout, and fully connected layer followed by a softmax layer. The classifier classified the videos either as real or fake.

The deepfakes videos and images generated using generative techniques are tedious to detect. To detect such manipulations, authors designed a multi-scale module consisting of CNN and ViT transformer for determining manipulations in facial regions [114]. Face details were captured using dilation convolution and depthwise separable convolution. ViT captured the global feature information using multi-head self-attention and utilized MLP for image classification as fake or real. Similarly, authors utilized MLP for learning the classification parameters for deepfake and real [113]. CNN extracted the facial landmarks consisting of temporal and shape features. The outcomes from the CNN and MLP concatenated using a fully connected layer with ReLU activation and a dense layer with a sigmoid activation function. In [88], authors utilized pre-trained model weights using a weight transfer technique for multi-class classification in the ViT framework. ViT comprised of attention mechanism, global content, and classification module for determining manipulations and alteration in low-quality datasets. On the other hand, Amin et al. [66] exploited a multi-transformer comprised of a SpectraVisionFusion transformer. The framework extracted features in the spatial and frequency domain from two different networks in the encoder. Weighted feature fusion mechanisms were embedded to obtain fused feature vectors for performing the classification from the transformer decoder and classification head layer.

Table 6
Single modal advanced deepfake detection techniques with deep learning-based algorithms.

SN	Citation	Features	Modality	Technique	Dataset	Performance metrics	Summary
1.	Khalid et al. [34]	Deep features from convolutional layers	Video	Graph NN	FaceForensics++, Celeb-DF, DFDC and World leaders DB	ACC and AUC	• Determined features with smaller spatial sizes in the image frames. • Constructed graphs by extracting edge information using K nearest neighbor. • Cross-DB analysis to prove the generalizability of the method.
2.	Nguyen et al. [35]	3D convolution kernels for extracting spatio-temporal features	Video	3D CNN	FaceForensics++ and VidTIMIT	ACC	• Extracted spatial and temporal features from consecutive video frames. • Utilized complex network models and different kernel sizes for network evaluation. • Achieved highest ACC (99.7%) on VidTIMIT DB.
3.	Raza et al. [97]	Spatial, temporal and spatio-temporal features using transformer	Video	Convolutional transformer	FaceForensics++, Celeb-DF, and DFDC	ACC, AUC and F1-score	• Combined different views of a video for analyzing deepfakes. • Extracted suspected regions by extracting different features. • Lightweight and fast model in comparison to existing state-of-the-art with fewer hyperparameters.
4.	Lin et al. [114]	Dilation convolution and depthwise separable convolution	Image	ViT	Celeb-DF, DFDC, FaceForensics++, and WildDeepfake	ACC and AUC	• Extracted spatial domain information from facial regions. • Integrated dual-subnet and multi-scale information for effective deepfake detection. • Captured facial regions at multiple scales for enhancing model's detection performance.
5.	Zhu et al. [27]	Spatial frequency features	Video	ST-Transformer	FaceForensics++, Celeb-DF, DFDC, DeeperForensics, and FaceShifter	ACC and AUC	• Adopted two-stage spatiotemporal encoder for handling inconsistencies in highly compressed videos. • Utilized knowledge distillation to represent structural information for deepfake detection. • Temporal contrastive module to enhance temporal correlation information from videos.
6.	Liu et al. [76]	Channel specific, channel differential and channel invariant features	Audio	BiLSTM	ASVspoof2019, ASVspoof2021, and VSDC	ACC, PR, RE, F1-score, and error rate	• Exploited multi-space channel representation learning for detecting audio deepfakes. • Channel differential encoder for extracting features by calculating difference between two channels. • Utilized orthogonal and mutual maximization loss information for channel-specific encoders.
7.	Amin et al. [66]	CNN encoder based facial feature extraction	Video	SpectraVisionFusion Transformer	FaceForensics+, DFDC, Celeb-DF-v2, and DeeperForensics-1.0	ACC, AUC and PR	• Extracted spatial, frequency and statistical features for improved detection results. • Weighted features embedding fusion for combining results from different networks. • Squeezed convolution projection in the multistage convolutional token embedding layer.
8.	Gowrisankar and Thing [6]	Saliency feature map using XceptionNet	Video	MesoNet	FaceForensics++ and Celeb-D	ACC and computational time	• Selected and ranked important facial regions for deepfake detection. • Generated adversarial video samples using NES with minimal noise difference from the original videos. • Utilized multiple explainability tools to describe model's black box design.

(*continued on next page*)

Table 6 (*continued*).

9.	Pu et al. [36]	Feature level and frame-level features	Video	ResNet50	Celeb-DF (v2), Celeb-DF, FaceForensics++ and DFDC	ACC, AUC, PR, RE and F1-score	• Handled data imbalance by using sampling and data augmentation techniques. • Utilized loss function to minimize the AUC error and cross-entropy loss. • Two separate classifiers for classifying the feature level and frame level outcomes.
10.	Kolagati et al. [113]	CNN based feature extraction	Video	MLP	DFDC and Dessa DB	ACC and AUC	• CNN based feature extraction for eye blink and lip, eye and nose shape. • Utilized two-layered MLP architecture with dense and hidden layer ReLU activation function. • Concatenated MLP and CNN output using fully connected dense layer followed by Sigmoid activation function.
11.	Arshed et al. [88]	Pre-trained weights from ImageNet DB	Image	ViT	Kaggle DB, Online DB, SFHQ, and SG DB	ACC, PR, RE, F1-score, and support	• Focused on multi-class classification of deepfake images. • Curated SG DB for supporting multi-class classification for better accuracy. • Achieved detection ACC up to 99% during model validation and testing.
12.	Awotunde et al. [95]	Discriminant spatial features from DL model	Video	ReLU with CNN	DeepFake DB, Face2Face DB, and First order motion DB	ACC, RE, PR, F-Measure, error, SPE and CM	• Utilized CNN for extracting face regions from video frames. • Enhanced CNN with ReLU for extracting spatial information used in identification of face artifacts. • Fully connected layer with neural network regularization for model robustness.
13.	Ayman et al. [69]	CNN based feature map	Video	VGG-16	DFDC	ACC	• Proposed deepfake generation and detection to understand the forgery regions. • Binary cross entropy to determine the percentage of fake and real production. • Achieved highest ACC (98%) in CNN.
14.	Badale et al. [96]	Frame level features using dense CNN	Image	CNN	SG	ACC	• Detected pixel-level manipulations using CNN. • Computed multiple ACC using different loss functions and optimizers.
15.	Coccomini et al. [40]	Hierarchical feature representation with low-dimensional features using EfficientNet-V2 CNN	Video	ViT, and Swim transformer	Pristine DB, and SG	ACC, CM, and AUC	• Compared capabilities of multiple DL models for detecting deepfakes using two different steps. • ViT shown superiority in comparison to other models for generalizability. • Inferred that scenario-specific DL-models perform better.
16.	Guo et al. [89]	Hierarchical feature extraction for high-level features	Image	AMTENnet	FaceForensics++ and SG hybrid fake face DB	ACC, CM and RER	• Learned manipulation traces adaptively using backpropagation method. • Integrated AMTN with CNN for robust model representation. • Applied post-processing techniques for simulating model under complex scenarios.

(*continued on next page*)

To provide a generalization set up for deepfakes detection, authors exploited three network architectures namely, CNN, ViT, and swim transformer [40]. The performance of these network architectures was analyzed and compared individually. CNN utilized EfficientNetV2-M and swim transformer to transform the hierarchical features into low-level features with attentional mechanisms. CNN showed efficient performance on limited training datasets whereas ViT showed superior results on large and diversified datasets. The swim transformer showed context-based generalized and promising results in both scenarios. Guo et al. [89] proposed adaptive manipulation traces using a convolutional

Table 6 (*continued*).

17.	Mcuba et al. [7]	MFCC, Mel-spectrum, Chromagram, and spectrogram	Audio	Custom DL architecture	Baidu Silicon Valley AI Lab DB	ACC	• Used different optimizers with variable learning rate to determine model accuracy. • Custom model achieved 100% ACC in complex scenarios. • Smaller DB size restricted the generalizability of the model analysis.
18.	Lima et al. [75]	Spatio-temporal features	Video	Recurrent CNN, ResNet-3D, C3D, ResNet Mixed 3D-2D, ResNet (2+1)D and I3D	Celeb-DF V2	ACC, and AUC	• Applied random cropping and temporal jittering to all video frames for extracting important information. • Exploited benefits of temporal features for deepfake classification. • R3D achieved highest ACC (98.26%) in comparison to other networks.
19.	Yan et al. [61]	Global features	Video	Convolutional autoencoder transformer	UADFV, FaceForensics++, Celeb-DF v2, DFDC, and DeepfakeTIMIT	AUC and CM	• Optimized encoder using back propagated information from various tasks. • Utilized latent vectors in shallow linear network for prediction. • Verified model generalizability with cross-domain validation.
20.	Wodajo et al. [65]	CNN based feature extraction	Video	Convolution ViT	DFDC	ACC	• Combined the capabilities of both CNN and transformers. • Local features learned from CNN whereas global and local features learned from ViT. • Model achieved ACC (91.5%) on test dataset.

DB: Dataset DFDC: Deepfake Detection Challenge LSTM: Long Short-Term Memory AMTE: Adaptive Manipulation Traces Extraction SFHQ: Synthetic Faces High Quality ViT: Vision Transformer ReLU: Rectified Linear Unit MLP: Multilayer Perceptron CNN: Convolutional Neural Network ACC: Accuracy AUC: Area Under the Curve PR: Precision RE: Recall SPE: Specificity CM: Confusion Matrix RER: Relative Error Reduction.

layer for generalized fake face detection architecture. Maxpool, ReLU, layers, and batch normalization were utilized after each convolutional layer for introducing non-linearity and preventing model over-fitting. The classification network consisted of three fully connected layers for determining whether facial images as real or manipulated. In [96], the author extracted features using CNN by applying a 2D convolution operation on 9 pixels of image regions. The prediction was done by applying a classification model to determine the class of the image. On the other hand, Awotunde et al. [95] proposed a five-layered CNN for detecting deepfakes. Enhanced CNN with ReLU for extracting the robust features from the facial region. The classification model comprised of ADAM optimizer and mean square error to calculate the loss and improve model stability and accuracy.

In another direction of research, Khalid et al. [34] utilized a graph neural network for deepfake detection. The network consisted of two stages namely GraphNet for updating and aggregating graph information, and FFN for linear transformation of the nodes. PReLU activation function for improving model performance and addressing vanishing gradient problem. The author proposed to generate a saliency map by segmenting the facial regions [6]. XceptionNet and MesoNet were utilized for training models and feature extraction, respectively. Explainability tools interpreted the model architecture to estimate the parameters for the adversarial generated sample. In [69], authors generated feature maps using CNN and processed the extracted features using VGG-16. CNN utilized a pooling layer for downsampling to reduce the network parameters and a fully connected layer for final video classification as real or fake. Authors extracted spatial and temporal features for deepfake detection [75].

Detecting deepfake audio is challenging as the cloned audios learn similar features and characteristics that are tough to recognize. To address this, Mcuba et al. [7] extracted audio features and represented them using visual features namely, MFCC, Mel-spectrum, chromatogram, and spectrogram. The performance of multiple DL-algorithms was analyzed and compared. It was observed that VGG16 performed better for MFCC audio features whereas custom architectures performed best for the rest of the features. However, authors proposed to detect deepfakes audio by representing the channel-specific, channel invariant, and channel differential features in multi-space channel [76]. The differences and commonalities between the two channels namely, the left and right audio were analyzed and classified using BiLSTM and a fully connected layer. A dynamic weighted fusion was adopted for generating joint features in a multi-space feature aggregation module before being fed to the ADD classifier. This classifier consisted of 1D-CNN, a fully connected layer, and sigmoid activation for predicting fake speech. Spatial features were extracted using CNN whereas LSTM was for extracting temporal features. These features were processed using 3D CNN, ResNet (2+1)D, Inflated inception V1, and RCNN. The results from these networks were compared to obtain an efficient network with better accuracy.

To summarize, DL-based models for deepfake detection demonstrated satisfactory performance by extracting deep features. These features were processed either using DL-models or transformers. Pretrained models are utilized for performing feature evaluation to improve the computational efficiency of the model. ViT transformer is one of the efficient networks consisting of an encoder layer and a decoding layer. The decoding layer is followed by a classification network for classifying the audio, video, and images as manipulated or real. Apart from this, Swim transformers have also shown efficient and accurate classification accuracy. The model efficiently processed and classified the various multimedia content and detected the discrepancy efficiently.

5.2. Hybrid fusion-based algorithms

Table 7 tabulates the representative work categorized as hybrid algorithms for deepfake detection. These algorithms utilized multiple

DL-model for feature extraction and processing. The classification module can utilize either the DL model or ML model for classifying various contents as real or fake. In this direction, Dincer et al. [37] utilized three pre-trained DL-models for feature extraction using transfer learning and two capsule networks for feature processing and classification. A frame selection strategy was adopted to select the suitable feature frame for feature extraction to improve detection performance and processing time. The classification scores from each capsule network were integrated using the mean fusion strategy to obtain a fused score for classification decision-making. Authors proposed hybrid CNN for extracting frame-level features from the selected video [70]. Hybrid CNN comprised two DL networks namely, InceptionResNet v2 with 22 convolutional layers, and 5 pooling layers, and Xception with depth-wise separable convolutional layers. The InceptionResNet v2 was followed by Xception network for feature extraction from pre-processed videos and then the classification module generated classification results. Similarly, the author ensembled two pretrained CNNs and a bidirectional LSTM network for detecting deepfakes [80]. Frame-level features were extracted using a customized network consisting of InceptionResNet v1 and InceptionResNet v2. These features trained the BiLSTM for the classification of deepfake videos using the majority voting rule. The network performance was optimized using bio-inspired spotted hyena.

Highly realistic fake videos that are hard to detect can be detected by adopting hybrid networks. For this, the authors proposed a hybrid approach by integrating you only look once (YOLO) v3, CNN, and XGBoost [62]. YOLO detected the facial regions from the video frames. Spatial features were extracted using pre-trained InceptionResNet v2 CNN. The classifier module was constructed using an ML-based XGBoost algorithm. Naskar et al. [68] presented a staking-based ensemble approach in which features were extracted using two DL networks namely, XceptionNet and EfficientNet-B7. Ranking based feature selection strategy was adopted to select the top-performing features. The classification module utilized a meta-learning approach based on MLP algorithms capturing non-linear relationships between features for improved classification accuracy. Xu et al. [48] proposed a set CNN for detecting facial manipulated videos using XceptionNet and MesoNet as backbone networks. Set reduce was utilized for aggregating the frame level and set level features. Multi-way backbone network branches were concatenated using a fusion module before classification.

In order to improve the detection performance in low-quality images, authors proposed to transform extracted features into higher-quality images using U-net [117]. Features were extracted using modified EfficientNet and adversarial training along with a discriminator was adopted for determining the quality of transformed features. The classification module was constructed using CNN consisting of convolution and a fully connected layer followed by softmax activation. The authors analyzed the local and global context of the video using three transformers namely, dual attention ViT, Inception, and group propagation ViT [38]. Dual attention ViT extracted contextual information, the frequency spectrum was broadened using an Inception transformer, and high-resolution features were retained using group propagation ViT. The outcome of these transformers was fused using an MLP mixer and then passed through a dense layer for classification. Qadir et al. [24] proposed a hybrid network consisting of three networks as ResNet-Swish-Bidirectional LSTM. Spatiotemporal data was processed by integrating bidirectional LSTM in ResNet-18 architecture. The swish activation method optimized the network and improved its feature selection and recall capabilities.

In [116], authors proposed a hybrid model for deepfake detection extracting face texture and naturalness degree. The texture was extracted using CNN and the naturalness degree was built using deep neural networks. Critical 68 landmarks were identified from the facial regions to extract these features. The model concatenated the outcome of all neurons from the naturalness degree model and used this for classification. Raza et al. [63] presented a hybrid approach of CNN-VGG16 along with transfer learning architecture for deepfake detection. Transfer learning architecture such as Xception, Neural search architecture network, MobileNet, and VGG16 models for feature extraction. CNN was involved in extracting pixel-level information and fully connected layers were combined to classify the outcomes. Authors stacked multiple DL models such as MobileNet, ResNet101, Inception v3, DenseNet121, InceptionResNet v2, DenseNet169, and Xception using ImageNet weights for deep fake classification [86]. The outcome of these models was concatenated and passed through CNN based classifier for making predictions. On the other hand, the authors combined the strength of CNN and GAN in a two-step approach for obtaining efficient prediction results [25]. Initially, GAN identified the discriminative parts in the synthetic images followed by the CNN model to improve the classification accuracy for forged images. GAN-based discriminator consisted of five convolutional layers and a LeakyReLU activation function whereas CNN had two convolutional layers and a ReLU activation function. Fully connected layers acted as a classifier for obtaining the outcome. Similarly, Sharma et al. [98] ensembled CNN-GAN to determine the deepfake generated using generative modeling. GAN was trained using real image datasets whereas CNN was trained on merged datasets. After this, the DCGAN network was adopted consisting of a generator and discriminator along with cross entropy for distinguishing between fake and real images.

Automatic reliable detection of deepfakes is essential to predict the forged content quickly. In this direction, Zhang et al. [91] ensembled 3D CNN and CNN-RNN (Recurrent neural network) for extracting discriminative spatial–temporal features and processing them for efficient deepfake detection. The network performance was optimized using PSO (Particle swarm optimization). A weighted scheme was adopted for integrating the outcome from these networks and classifiers were selected for classifying the outcome. Authors segmented the various facial regions (68 landmarks) for extracting the features using OpenFace2 [60]. The multiple detection models namely, CNN, capsule network, and DSP-FWA were utilized for processing these features. These features were concatenated using two different fusion approaches namely, model fusion and feature fusion. The fusion approach was based on weighted fusion to obtain the final score fed to the classifier. In [9], authors utilized two networks namely CNN and LSTM for extracting spatial and temporal features. These features were exported to the training model and prediction was done on the testing model. Authors ensembled CNN models with SVM and KNN for obtaining the accurate classification accuracy [100]. Error level analysis was performed to determine the compression level of the processed image. Features were extracted using transfer learning from two CNN models namely AlexNet and ShuffleNet. The features were fed to the ML classifiers namely, SVM and KNN for obtaining the accuracy values.

Determining the specific regions that were manipulated is challenging and requires in-depth analysis. To analyze such regions, authors studied the boundaries of the input data and processed and classified them using pre-trained DL model [115]. Acoustic features from frame-level boundaries were extracted using ResNet-1D and 1D-CNN networks. These features were classified using a backend classifier consisting of a transformer encoder and bidirectional LSTM. In other directions, Khochare et al. [67] ensembled DL models extracting image features and ML models extracting audio features for deepfake detection. Mel spectrogram images were generated from audio signals using two DL models namely, temporal CNN and spatial transformer network. Temporal CNN was used in classification networks with varying kernel sizes, dropout, and batch normalization. For feature-based classification, 37 audio features were extracted and classified using ML algorithms namely, SVM, KNN, RF, XGBoost, and Light GB. DL-based models had shown better performance in comparison to ML-based models. To maintain the interpretability of the model for detecting deepfake audio, authors used three DL-models namely, CNN, CNN-LSTM, and CNN-LSTM-permuted [122]. The audio data was converted to mel spectrogram images and features were extracted using CNN.

Table 7
Single modal advanced fusion-based deepfake detection techniques with hybrid algorithms.

SN	Citation	Features	Modality	Fusion Strategy	Dataset	Performance metrics	Summary
1.	Essa [38]	Low frequency and high-resolution group features	Video	Dual attention ViT, Group propagation ViT, and Inception transformer	FaceForensics++ and Celeb-DF	ACC, PR, RE, F1-score, and AUC	• Comprehensive analysis of videos using multiple ViT networks. • Integrated features using MLP network to improve detection capabilities. • Achieved highest ACC (97.72%) on benchmark DB.
2.	Cai and Li [115]	ResNet-1D for frame-level features	Audio	Transformer and BiLSTM	ADD2022-T2, ADD2023-T2, and PartialSpoof	Error rate and F1-score	• Extracted diverse deep features from pre-trained models. • Integrated results from multiple models for determining spoofed audio. • Limited performance to generalize results on unseen and cross-domain DB.
3.	Jayashre and Amsaprabhaa [80]	Deep features	Video	InceptionResNetV1, InceptionResNetV2, and BiLSTM	Kaggle DB, FaceForensics++, and FakeAVCeleb	ACC, PR, RE, F1-score, Loss and AUC	• Obtained fused feature by integrating hidden features extracted from InceptionResNetV1 and InceptionResNetV2. • Optimized network weights using spotted hyena optimizer. • Utilized majority voting for classifying videos as fake or real.
4.	Dincer et al. [37]	Feature extraction using VGG19, EfficientNet B0, and EfficientNet B4	Video	CapsuleNet and ArCapsNet	Celeb-DF and DFDC-P	ACC, PR, RE, F1-score, and AUC	• Frame selection by investigating golden ratio from the facial regions. • Fused features extracted from multiple deep neural networks. • Exploited two capsule networks in the classification models for detecting deepfakes.
5.	Khochare et al. [67]	Feature-level extraction and image-level extraction	Audio	SVM, RF, XGBoost, KNN, LGBM, STN and TCN	Fake or Real DB and self-generated audios	ACC, PR, RE, F1-score	• Implemented feature-level extraction and image-level extraction for detecting deepfake audios. • Feature-based approach utilized ML algorithms and image-level approach used DL methods. • SVM achieved ACC (67%) and TCN acquired ACC (92%).
6.	Chew et al. [116]	Texture and naturalness degree	Image	Deep CNN	Flickr-Faces-HQ, StyleGan2, FaceForensics++, and Celeb-DFv2	ACC, PR, RE, and F1-score	• Utilized hybrid model for extracting deep texture and naturalness degree from facial regions. • Achieved highest ACC (99.7%) on benchmark DB.
7.	Zhang et al. [91]	3D CNN extracting spatial and temporal features	Video	PSO optimized CNN-RNN and 3D CNN	Celeb-DFv2, FaceForensics++, and Deepfakes	ACC, MCC, and AUC	• Exploited Inceptionv3-RNN network as encoder–decoder architecture. • Inflated 3D ConvNet and MC3 networks as 3D CNN for feature extraction. • PSO optimized the network hyperparameters for video authenticity.
8.	Ke and Wang [117]	Deep features using modified EfficientNet	Images	U-net and CNN	DFDC, Celeb-DF, FaceForensics++, and Wild Deepfake	ACC	• Utilized transforming module to improve the quality of the features extracted from degraded quality images. • Assessed upgraded quality of the images using a discriminator. • Utilized ML algorithms such as SVM and Bayes classifier for classifying the input data.

(*continued on next page*)

These features were processed using the LSTM network and attribution scores were calculated using integrated gradients. Model explainability was achieved using Taylor decomposition and speech was reconstructed using Griffin-Lim algorithms. Model detection performance was calibrated on multiple audio deepfake datasets.

To summarize, hybrid approaches utilized multiple DL algorithms and transformer networks for extracting various frame-level and facial region features. The boundary level features were also analyzed to

Table 7 (*continued*).

9.	Naskar et al. [68]	Dual CNN models	Videos	Ensembled Xception and EfficientNet-B7	FaceForensics++ and Celeb-DF V2	ACC, PR, RE, AUC, F1-score, and CM	• Adopted feature selection strategy by averaging the feature importance scores. • Arranged features in descending order to select the valuable features. • Meta learning applied on stacked features along with MLP.
10.	Qadir et al. [24]	Feature extraction from ResNet layers	Video	ResNet-Swish-BiLSTM	FaceForensics++, Celeb-DF and DFDC	ACC, PR, RE, AUC, F1-score, and CM	• Examined cross-compilation check among DB to determine model generalizability. • Hybrid and optimized DL architecture based on BiLSTM and residual network for robust model architecture. • Visual features in neurons to prevent the model from cyber-attacks.
11.	Raza et al. [63]	Learned features from TL	Image	Hybrid VGG and CNN	DF DB from Kaggle	ACC, Loss, PR, RE, SPE, F1-score, CM, and GME	• TL-based models for comparing the outcomes of the proposed model. • Tuned hyperparameters to determine fine-tune values. • Computed performance on unseen DB from the 10% testing DB.
12.	Ikram et al. [70]	InceptionRes-NetV2 CNN and Xception	Video	Hybrid CNN	DFDC	ACC, PR, RE, AUC, F1-score, and support	• Utilized InceptionResNetV2 and Xception in hybrid CNN model. • Global average pooling and softmax layer in the final processing layer. • Hybrid model achieved validation ACC (93%).
13.	Ismail et al. [62]	InceptionRes-NetV2 CNN	Video	YOLO-CNN-XGBoost	CelebDF-FaceForensics++	ACC, AUC, PR, RE, SPE, SEN, F1-measure, and CM	• Spatial features extracted from CNN network classified using XGBoost. • Stochastic gradient for optimizing the model for faster convergence. • Validated the model generalizability with cross-domain DB.
14.	Rafique et al. [100]	Deep CNN features	Image	AlexNet, ShuffleNet, KNN, and SVM	Computational Intelligence Photography Lab, Yonsei University	ACC, RE, PR, and F1-score	• Error level analysis to compare the compression of fake and real images. • Utilized DL-models for feature extraction and ML for classification into real and fake. • Achieved highest ACC (88.2%) in hybrid ShuffleNet and KNN model.
15.	Rana and Sung [86]	Seven DL models	Video	CNN as meta learner classification	FaceForensics++	ACC, AUC, PR, RE, F1-score, SPE, and SEN	• Ensembled multiple DL-models as base learners for feature extraction. • Utilized DeepFakeStackClassifier as meta learner before final prediction. • Achieved ACC (99.6%) on exploited public dataset.
16.	Sadhna et al. [25]	Hierarchical feature	Image	CNN and GAN	Public DB	ACC, Loss, and CM	• Compared between real and fake images along with machine-generated fake images. • Hybrid model achieved ACC (94.4%).
17.	Sharma et al. [98]	Deep features	Image	CNN and GAN	DFDC	ACC, AUC, PR, RE, F1-score, support, and CM	• Utilized transpose convolutional to reduce the features dimensions. • Randomly cropped images normalized using batch normalization. • Achieved ACC (98.67%) on Kaggle DB.

(*continued on next page*)

Table 7 (*continued*).

18.	Tolosana et al. [60]	Deep features from facial regions	Video	Xception, Capsule network, and DSP-FWA	UADFV, FaceForensics++, Celeb-DF, and DFDC	ACC, AUC, and error rate	• Extracted deep features from facial regions such as eyes, nose, mouth, etc. • Fusion classification model using multiple DL models and face-level. • Achieved 100% AUC on UADFV DB.
19.	Vamsi et al. [9]	ResNext CNN for frame-level features	Video	ResNext CNN + LSTM	Celeb-DFv2	AUC and CM	• Feature extraction using ResNet and temporal analysis using LSTM. • Achieved average ACC (91%).
20.	Xu et al. [48]	XceptionNet	Video	XceptionNet and MesoNet	TIMIT, FaceForensics++, and DFDC	ACC, and AUC	• Utilized set reduce method for aggregating features of elements in a set. • Statistical features for characterizing permutation invariance. • Processed multiple video frames as a set for detecting manipulated videos.
21.	Lim et al. [122]	CNN-based spectrogram-based features	Audio	CNN-LSTM-perm	ASVspoof2021 Logical Access DB, and LJSpeech DB	Attribution score	• Interpreted attribution score of audios to identify deepfake. • Focused on explainable model rather than model performance. • Utilized Griffin-Lim algorithm for phase reconstruction-rule based vocoders.

DB: Dataset, DL: Deep learning, ML: Machine learning, ViT: Vision transformer, LSTM: Long short-term memory, MLP: Multilayer perceptron, DFDC: Deepfake detection challenge, CNN: Convolutional neural network, RNN: Recurrent neural network, KNN: k-Nearest neighbor, SVM: Support vector machine, RF: Random forest, LGBM: Light gradient boosting machines, TCN: Temporal convolutional network, STN: Spatial transformer network, RNN: Recurrent neural network, PSO: Particle swarm optimization, GRU: Gated recurrent unit, BiLSTM: Bidirectional LSTM, TL: Transfer learning, ACC: Accuracy, PR: Precision, RE: Recall, GMS: Geometric mean score, MCC: Mean convergence curve, AUC: Area under the curve, CM: Confusion matrix, SPE: Specificity, SEN: Sensitivity.

determine the manipulation in specific regions. The extracted features were processed and classified using DL-models or ML-models. Feature selection strategy was adopted to maintain the computational complexity of the model. Model explainability and interpretability were explored to determine the difference between the model prediction and human perception.

6. Multimodal deepfake detection

Multimodal deepfake detection emphasizes incorporating multiple modalities such as audio, video, images, and text in the input features for detecting manipulation in practical scenarios. Multimodal deepfake detection techniques can be categorized either as DL-based approaches or ensembled approaches. Synthetic-generated multimedia can be detected using DL architectures, transformers, and integrating the outcome using fusion approaches for classification. The generic architecture of multimodal deepfake detection algorithms is illustrated in Fig. 9. Table 8 tabulates the representative work categorized under this class of deepfake detection algorithms. The next section will detail the DL-models for deepfake detection.

6.1. Deep learning-based techniques

This subsection will detail the multi-modal deepfake detection algorithms in which DL-model was exploited for feature detection, processing, and classification. In this direction, Yoon et al. [50] processed triple modality namely, text, video, and audio using a multimodal transformer. The transformer comprised of two stages including feature extraction and representation along with residual connection and late fusion. The visual feature loss due to the interaction of three modalities was prevented using the ReLU activation function. The distillation tokens and multimodality data scores were extracted and fused to obtain classifier probability for detecting deepfakes. In [49], authors extracted features from multimodality, audio, and video and fused them using the Intramodality mixer layer and Intermodality mixer layer. Audio features were sampled and learned using residual blocks whereas multiple residual and pooling layers were used for extracting video frames. The mixed features were passed to multilabel prediction heads to learn patches for classification as real or fake.

To prevent the uncertainties in models trained on fused multimodal data, authors proposed cross-modality and within-modality regularization learning [74]. Audio and visual were extracted separately using frame-level encoders and concatenated using AV-Hubert. Contrastive learning along with cosine similarity to determine the similarities between the audio and visual features. Within modality, regularization was achieved using modality-specific regularization consisting of margin-based regularization. Weighted cross entropy loss to optimize loss and detect deepfakes. On the other hand, authors generated intra-modal consistency loss and cross-modal consistency loss for detecting deepfakes [29]. Intra-modality loss determined the divergence between the feature vectors from different modalities and generated similarity scores. Whereas cross-modality extracted motion inconsistencies learned various feature extractors and generated similarity scores. These two scores were summed to obtain the final score for deepfake detection.

In [90], authors exploited pre-trained fine-tuned ViT for extracting video clips and audio spectrum from audio and visual data. Two additional modules adaptive forgery-aware injection and audio-distilled cross-modal interaction were stacked in each ViT for extracting intra-modal embeddings. The results were fed to the MLP-based classifier to detect the fake or real samples. On the other hand, authors proposed ViT with Xception network for learning the consistency in video and images from the entire facial regions [39]. The local features were extracted by combining ViT and self-attention module whereas global spatial features were extracted using deep CNN. These features were stacked, fused, and classified using a classification layer consisting of the ReLU activation function to generate the probability of classifying it as fake or real. Ilyas et al. [83] extracted hierarchical features using the Swim transformer module for representing the local and contextual information from the video and audio modalities. The consecutive Swim

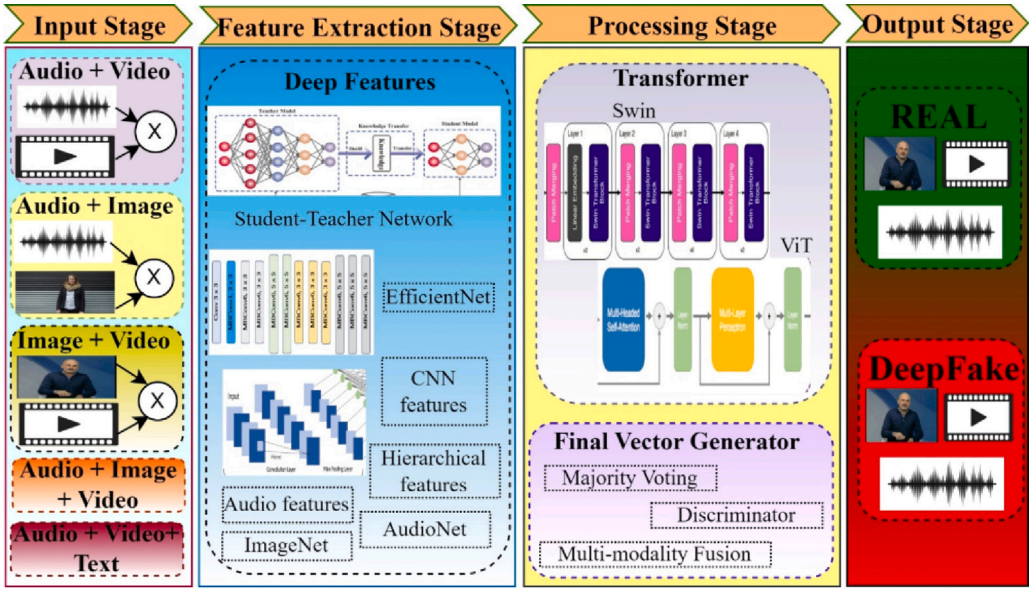


Fig. 9. Generic architecture of multi-modal deepfake detection algorithms.

transformer blocks represented the ST network comprising of layer normalization, multi-head self-attention module, residual network, and MLP module. MLP was four layered modules consisting of two identical layers, ReLU and dropout along with a dense layer for classification.

Also, the authors proposed an effective method that can detect multimodalities, audio, and videos together as well as can detect them individually in low-quality and corrupted videos [123]. The person of interest was identified, and discriminative features were learned using contrastive learning. The similarity indices were computed and fused linearly by averaging to obtain the fused scores for classification. Haq et al. [99] extracted emotions and psychological features for determining the inter and intra inconsistencies between the multi-modalities namely, audio and video. Emotions were identified and extracted using a DL network with multiple convolutional layers having ReLU, max pooling, and batch normalization. Speech emotions were recognized with seven basic emotions by tuning hyperparameters. Intra-modality reasoning computed the probability for two constitutive emotional states whereas inter-modality reasoning was determined by representing it in the arousal valence dimension. These modalities were classified by using probability labels for each reasoning module and explained using Grad-CAM. In [93], authors utilized the Xception network along with bidirectional LSTM for extracting multi-frame features. Audio waveforms were processed using SincNet and the detection task was processed using the final classifier. After this, a discrete Fourier transform was applied on images to extract the power spectrum for separating the fake from the real using a trained MLP classifier.

To summarize, pre-trained DL networks and transformers were used for extracting the robust features from the considered multi-modalities. The intra-domain and inter-domain modalities were processed to determine the inconsistencies between the modalities. The modalities score was fused using DL-based or MLP classifier for the classification of modalities as fake or real.

6.2. Ensembled fusion-based techniques

These techniques utilized multiple DL-models to extract robust features and multiple fusion modules to effectively utilize the extracted information. The results were ensembled to generate the final robust score for deepfake classification. In this direction, Yan et al. [53] utilized two GNN (Graph neural networks) to extract geometric enhanced landmark features namely, spatial, frequency, and temporal. Multi-scale Xception network was used for extracting spatial features. Temporal information was encoded and inconsistencies between frames were determined by using a graph attention network at the video level. Unified features were generated from different frequencies and fused using a learnable weight matrix. Global and local temporal dependencies at the video level from each attentional module were concatenated using a weighted sum of attentional weights from each node feature. Cross-modal classifier along with loss functions was concatenated to reduce the inter-class noise for deepfakes detection [118]. The temporal and spatial encoders processed the sampled audio and video through patch data processing. The multimodal decoder fused the extracted features using their inherent relationship. The output of the multimodal decoder was computed through cross cross-modality classifier using contrastive, cross-entropy, and angular margin loss. These losses were fused linearly to obtain the final classifier outcome. Similarly, authors utilized two fusion strategies namely, feature fusion and score fusion before feeding to the four-class classification network [51]. The multi-task learning utilized fusing three loss functions namely, cross-entropy loss for audio, video, and four classes with different types of videos. Feature fusion generated a fused feature vector after extracting features from the capsule network and Swim transformer.

Further, multimodal contrastive learning was utilized for extracting intra-frame and cross-modal for determining forgery clues [41]. Three modalities namely, audio, video, and images were extracted along with audio and images extracted for the cross-modality gap. Intra-modal similarities from the ST network were predicted using Jensen–Shannon divergence. The contrastive loss enhanced feature representation by concatenating the audio with video information. Noise-based adaptive feature augmentation to improve the model generalizability capabilities. Authors extracted facial features from the VGG-16 network and audio from a transformer encoder with a self-attentional network [64]. The spatiotemporal dimensions were captured using tubelet embedding and simplified computing cross-attention between lip region and audio temporal synchronization. The classification module consisted of MLP to generate the final classification probability for real and fake.

In other directions, authors utilized different networks for extracting multimodalities and different fusion strategies to detect the deepfakes [52]. Visual modalities were extracted using modified ensembled EfficientNetB4 and audio modalities were extracted using pre-trained time-delay neural networks provided by SpeechBrain. The time consistency between both these feature vectors was recorded by extracting

Table 8
Multi-modal deepfake detection techniques.

SN	Citation	Features	Modality	Technique	Dataset	Performance metrics	Summary
1.	Liu et al. [41]	DL-based feature extraction	Audio, image and video	Multimodal contrastive learning	DF-TIMIT,DFDC, FakeAVCeleb, KoDF	AUC, ACC	• Adopted student–teacher network for extracting multi-modal features. • Utilized noise-based adaptive feature augmentation for better generalizability. • Calculated feature mapping using MLP.
2.	Yoon et al. [50]	Transformer for extracting vision and audio features	Vision, audio and text	TMI-Former	FakeAVCeleb	ACC, F1-score	• Triple modalities processed in two stages consisting of visual and auditory features. • Residual connections for handling the features from vision, audio, and text. • Integrated the extracted multi-modalities using late-level fusion.
3.	Yang et al. [118]	Temporal and spatial features	Audio and video	Multi-modal joint decoder	DefakeAVMiT, FakeAVCeleb, DFDC	AUC, ACC	• Generated DB by combining videos and audio from multiple different DB. • Jointly learned the features extracted from different encoders. • Cross-modal classifiers for identifying deepfake videos.
4.	Lewis et al. [82]	Color visual embedding, spectral embedding, LipNet, and audio features	Video and audio	LSTM	DFDC	ACC	• Analyzed misalignment between audio and videos for deepfake detection. • Utilized DCT for capturing spectral features in frequency domain. • Inconsistencies between DeepSpeech2 and LipNet features explored using contrastive loss.
5.	Ganguly et al. [39]	Deep CNN based features	Image and Video	ViT with XceptionNet	FaceForensics++, Celeb-DF (V2), Deepfake DB	ACC, AUC, PR, RE, F1-score, CM	• Patch-wise attentional module for learning local features. • Pre-trained Xception network for extracting global spatial image features. • Dense layer-based classification network for detecting deepfakes.
6.	Ilyas et al. [83]	Dense hierarchical features	Audio and video	Swin transformer	FakeAVCeleb, Celeb-DF, ASVSpoof-2019 LA, World Leaders DB, Presidential Deepfakes DB	ACC, AUC, PR, RE, F1-score, error rate	• Extracted deep hierarchical features for robust representation of model. • Adopted majority voting for classifying fake from real. • Fully connected layer along with softmax activation in the output block.
7.	Lomnitz et al. [93]	Xception	Audio and video	Ensembled model: EfficientNet, Xception, ResNet, SlowFast video-based network	DFDC	BCELoss	• Feature encoding using MTCNN and XceptionNet. • Frame-by-frame feature extraction from single to multiple frames. • MTCNN for ensembling different models for deepfake detection along with three 3D CNN.
8.	Nie et al. [90]	Task-specific features including audio and visual	Audio and Images	Forgery aware audio distilled multimodal learning	FakeAVCeleb, KoDF, DeAVMiT, DFDC	AUC, ACC	• Utilized learnable parameters from ViT transformer. • Audio-distilled cross-modal interaction for handling inconsistencies between audio and visuals. • Experimental analysis to prove the model scalability for superior detection outcome.

(*continued on next page*)

them for equally spaced time instants on the same number of frames. Before classification, the feature vectors were fused using three strategies namely, late, mid, and early. In late fusion, separate classifiers for each feature vector and vector were averaged to obtain the final score. Similarly, separate classifiers were used in the mid fusion and results were merged using the dense layer. However, a unique classifier was considered in late fusion, and concatenated feature vectors were fed to it to obtain the final detection results.

Table 8 (*continued*).

9.	Muppala et al. [51]	Capsule network and Swin transformer	Audio and video	Multi-task learning	FakeAVCeleb, TMC DB	AUC, ACC	• Extracted deep features to learn powerful features for model implementation. • Utilized three loss functions in multi-task model learning. • Four-class classification for detecting deepfakes.
10.	Kharel et al. [64]	VGG16	Audio and video	Self attention-based transformer	DFDC, DF-TIMIT, FakeAVCeleb DB	AUC, F1-score	• Analyzed lip synchronization between audio and visual artifacts. • Self-attention applied on video and cross-attention deployment for lip-video synchronization. • Final class labels for classification were generated by MLP.
11.	Haq et al. [99]	Pretrained facial expression recognition and speech emotion recognition	Audio and video	DL-based model	DFDC,WLD	ACC, CM	• Introduced biological signals to determine the emotion inconsistencies. • Analyzed intra-modality and inter-modality reasoning for robust outcomes. • Focused on explaining the model using DL-based explainability models.
12.	Cozzolino et al. [123]	Contrastive based feature learning	Audio and video	POI-based detection	pDFDC, DF-TIMIT, FakeAVCelebV2, KoDF	AUC, ACC	• Analyzed moving face and audio as discriminative identity for deepfake detection. • Robust model against low-quality or corrupted DB. • Evaluated and analyzed model flexibility, robustness, and generalizability.
13.	Zou et al. [74]	Frame-level audio-video features using encoders	Audio and video	Audio–visual transformer	FakeAVCeleb	AUC, ACC, PR, RE, F1-score	• Cross-modality regularization module for pairing audio-video signals and preserve modality distinctly. • Target-specific modality to retain modality information. • Cross-entropy loss function to optimize deepfake output.
14.	Yan et al. [53]	Geometric enhanced landmark features	Images and videos	Graph attention network	FaceForensic++, DeepfakeDetection, CelebDF	AUC, ACC, error	• Multiple modalities extracted using two GANs. • Highly complex model not suitable for real-time applications. • Generalizability proved using cross-dataset assessment but limited to considered modalities only.
15.	Tian et al. [29]	Consistency-based feature and joint features	Audio and video	Transformer	VoxCeleb2, FakeAVCeleb, KoDF	ACC, AUC	• Calculated cross-modal consistency loss from similarity matrix. • Intra and cross-modality scores compared for classification. • Model generalizability on multiple DB for adversarial attacks.
16.	Salvi et al. [52]	High level pixel-level features from video and audio	Audio and video	Time aware neural network	FaceForensics++, ASVspoof 2019, VidTIMIT, DeepfakeTIMIT, TIMIT- TTS	AUC, CM	• Time series audio and video features extracted for analysis. • Trained model on synthetic audio and video DB. • Evaluated model performance on unseen DB for model generalizability.

(*continued on next page*)

To summarize, ensembled approaches utilized different DL networks for extracting the different modalities of information from the datasets. In addition, multiple networks were also utilized for fusing the extracting information to generate final scores for classification. Different fusion strategies followed by classification module for generating the hierarchical robust features for determining the modalities as real or fake. However, certain concerns should be addressed to ensure the enhanced detection accuracy and robustness of the ensembled approaches. As this approach, integrates multiple DL models for improving the detection results, maintaining model computational complexity for real-time detection is a foremost concern. The number of overheads in terms of hyperparameters must be controlled to deal with the high computational requirements of the system. Reliable design and proper training of the model are crucial to prevent overfitting

Table 8 (*continued*).

17.	Raza and Malik [49]	ResNet-1D for audio features and multi-layered 3D Resnet for video features	Audio and video	Multimodaltrace model	FakeAVCeleb, PDD, WLD DB	ACC, PR, RE, F1-score	• Combined channel extractors and mixers for analyzing inter and intra modalities. • DL-based multi-label classification for deepfake detection in audio–visuals. • DB augmentation by incorporating other DB to increase training size.

DB: Dataset, DL: Deep learning, POI: Person of interest, DFDC: Deepfake detection challenge, KoDF: Korean deepfake DB, LSTM: Long short-term memory, ACC: Accuracy, BCE: Binary cross entropy, CM: Confusion matrix, AUC: Area under the curve, PR: Precision, RE: Recall, ViT: Vision transformer, MLP: Multi-layer perceptron.

and ensure model generalizability in practical situations. The decision-making process must be transparent and trustworthy by ensuring model interpretability and explainability for quick and fast detection of deepfakes. In addition, large and diverse multi-modal datasets must be provided to evaluate the performance and identify its limitations in deepfake detection.

7. Critical discussion

In this study, we have categorized deepfake detection algorithms as single-modal and multi-modal approaches based on the input modalities utilized in their model architecture. Single-modal architecture is focused on processing individual multimedia content namely, audio, video, or image only. Whereas multi-modal utilizes a combination of these modalities to generate a unified outcome for classifying the contents as fake or real. Deepfake detection comprises various steps that include feature extraction, feature processing, unified feature generation, fusion strategies, and classification modules. All the techniques under each category follow these steps but exploit single or multiple feature extraction algorithms, classifiers, and fusion algorithms.

Generally, conventional studies focus on extracting the handcrafted features and pre-trained DL models to reduce the computational complexity of the model in feature extraction [88]. The fine-tuning of the model is achieved by training the hyperparameters so that intrinsic aspects of the forged data can be captured efficiently. However, model efficiency is impacted when the performance is evaluated on unseen and real-time collected data. It is essential to compute the model with and without fine-tuning so that model effectiveness can be determined in real-time. Hybrid models combine several DL models for feature extraction and generate fused features for deepfake detection. The detection accuracy is improved but the model complexity needs to be addressed for faster processing of the content for deepfake detection.

Further, complex networks consisting of encoders, decoders along with residual networks are adopted for feature extraction and processing for deepfake detection. ViT, Swim transformer, and ST networks have shown great potential and adaptability for deepfake classification by extracting local and global features. In [65], authors have presented the ViT architecture along with CNN for detecting deepfakes with enhanced accuracy. The model consisted of two steps namely, feature learning and processing learned features into the sequence of image pixels for final detection. The feature learning involves seventeen convolutional layers, five max pooling, ReLU activation, and batch normalization. On the other hand, the authors utilized a ViT transformer with twelve transformer layers, 768 hidden layer components with GeLU activation function, and balanced datasets. Zhu et al. [27] utilized the ST network for extracting mutual information with temporal contrastive distillation. The model was trained on eight GPUs, consisting of four stacked encoders each with eight attention heads. To add to the model complexity, the classification head consisted of two fully connected layers. The deepfake detection models are highly complex and the processing performance with computational time is ignored in most of the studies.

For training deepfake detection models, a lot of benchmark datasets have been generated using DL techniques. The most popular and adopted datasets by most of the recent studies are UADFV [124], Deepfake-TIMIT [125], FaceForensics++ [126], DFDC (Deepfake detection challenge) [127], Celeb-DF v1, Celeb-DF v2 [128], DeeperForensics-1.0 [129], WildDeepfake [130] and many more [32, 131–133]. Apart from these visual datasets, there also exists benchmark datasets for predicting deepfake audio. Some latest audio deepfakes datasets are Asvspoof2019 [134], Arctic dataset [135], Css10 dataset [136] and VoxForge dataset [137]. Also, a multimodal dataset exists which consists of an amalgamation of visual and audio recordings [138]. Most of these datasets are either generated synthetically or noise is induced artificially to generate fake content. These datasets lack diversity and capture limited expression for deepfake detections. The authors introduced cross-dataset evaluation of the model to determine its capabilities in a variety of scenarios so that the model will not fail when implemented for real-time applications.

7.1. Benchmarking table

The proposed work is compared with existing studies in the area and the same is depicted by a benchmarking table (Table 9). Ten crucial and different attributes are identified and compared with eight other existing studies. The last row of the table represents the proposed study. The presence and absence of extracted unique and crucial attributes in the study are recorded by marking ‘’ and ‘’ in the table, respectively. It has been observed that only one study has followed the PRISMA strategy for article selection and elaborated the statistical distribution of the selected articles. Most of the studies discussed the deepfake detection algorithms without categorizing them to highlight the salient features. In addition, a limited number of studies were reviewed in their work ignoring the multimodal approach for deepfake detection. Most studies included feature extraction strategies for determining the spatial, and temporal features from the global and local information. The performance metrics for evaluating the model performance were elaborated by few of the studies. The strength of the proposed study is the categorization of deepfake detection algorithms as single-modal and multi-modal techniques. Further, salient features along with the advantages and disadvantages of each category for deepfake detection models provide future insights and trends to the researchers working in the domain. In addition, the adopted selection strategy, statistical distribution analysis along with elaboration of performance metrics further prove the robustness, efficiency, and potential of the proposed study.

7.2. A short note on generalizability and explainability in deepfakes

In order to ensure the modal performance in real-time and unseen situations, the generalizability of the model plays a crucial role. In addition, various DL-modal such as CNN, XceptionNet, ResNet, and many others with large numbers of convolutional layers, hidden layers, and filters were utilized for detecting deepfakes [139]. The prediction outcome of these complex models is opaque and uninterpretable. For a better understanding and interpretability of the model, it is essential to explore explainability in these models [140–143].

Table 9
Benchmarking table for comparing the proposed work with existing studies.

SN	Citations	Year[a]	PRISMA[b]	DFC[c]	FE[d]	CDF[e]	EDF[f]	MDF[g]	#SDFD[h]	PMA[i]	SA[j]
1.	Albahar and Almaki [56]	2019	×	×	×	×	×	×	15	×	×
2.	Seow et al. [2]	2022	×	✓	×	✓	×	✓	60	×	×
3.	Rana et al. [54]	2022	✓	✓	×	×	×	×	90	×	✓
4.	Malik et al. [57]	2022	×	×	✓	×	×	×	52	×	×
5.	Gupta et al. [58]	2023	×	×	✓	×	✓	×	67	×	×
6.	Mary and Edison [59]	2023	×	×	✓	×	×	×	20	✓	×
7.	Patel et al. [1]	2023	×	×	✓	×	✓	✓	68	×	×
8.	Passos et al. [55]	2024	×	✓	✓	×	×	×	101	✓	×
9.	Kumar et al. (the proposed)	2024	✓	✓	✓	✓	✓	✓	88	✓	✓

PRISMA[b]: Preferred Reporting Items for Systematic Reviews and Meta-Analysis
DFC[c]: Deepfake categorization
FE[d]: Feature extraction
CDF[e]: Conventional deepfake
EDF[f]: Ensembled deepfake Fusion
MDF[g]: Multimodal deepfakes
#STDF[h]: Number of deepfake detection studies
PMA[i]: Performance metric analysis SA[j]: Statistical analysis.

In deepfake algorithms, cross-dataset evaluation is performed to prove the model's generalizability. The cross-datasets analysis also addresses the limitations of small and less diverse dataset. Cross-datasets are challenging due to diverse, unseen, and realistic scenarios for training and testing the model performance [66]. The cross-dataset evaluation settings are also effective in exhibiting the model performance for inter-class and intra-class instances which are distinguishable for enhancing the model evaluation parameters for deepfake detection.

Next, explainability is one of the evolving technologies that provide a clear interpretation of DL networks by analyzing model architecture and parameters [144–147]. In deepfakes detection models, explainability provided model transparency by using decomposition and data perturbations as post-hoc approaches [122]. Layer-wise relevance propagation, Deep Taylor, and integrated gradients were utilized for evaluating the model sensitivity, attribution, and input relevance. The collected relevance score is approximated to obtain the heatmap prediction for model explainability. Authors achieved model design explainability by analyzing intra-modal and cross-modal consistency loss [29]. These losses can detect that portion of video which is inconsistent for detecting the video as real or fake. The model explainability in performing decisions was achieved using Grad-CAM (Gradient class activation mapping) [99]. Gradcam is an attentional mechanism that can highlight certain regions in an image or video by generating heatmaps for deepfake detection. Apart from Grad-CAM, other tools such as SOBOL, XRAI (eXplanation with ranked area integrals), RISE (Random input sampling explanations), and LIME (Local interpretable model agnostic explanations) were also observed for evaluating model performance for different perturbations [6]. These tools analyzed model's internal structure and worked based on the randomly selected input features.

To summarize, generalizability and explainability are two essential and crucial features that must be present in the design and architecture of deepfake detection models. These features ensure trust, transparency, robustness, and reliability in DL-based deepfake detection algorithms. An explainable model ensures a clear understanding and interpretability of the model decision-making process. On the other hand, generalizability ensures the model's suitability in practical scenarios. The presence of these two features in deepfakes detection algorithms embarks the transfer of common knowledge across domains.

7.3. Discussion on evaluation metrics for performance analysis

To ensure enhanced performance to detect disinformation in real-time, it is essential to utilize critical evaluation metrics to compute the performance of single-modal and multi-modal detection algorithms on public datasets. Exhaustive evaluation metrics such as accuracy, area under the curve (AUC), precision, recall, F1-score, confusion matrix, and error rate are exploited to determine model reliability and robustness to detect fake from real [39,64,99,123]. Most of the articles utilized accuracy and AUC to measure the in-depth robustness and reliability of the model to detect deepfakes correctly [28,30,81,118]. Accuracy computes the number of correctly detected instances with respect to the total number of instances provided to the model. The higher the accuracy better the model performance to detect fake from the real. AUC is another popular, scalar, and robust metric to compute the model's functional performance. It considers all classification thresholds to measure the receiver operating characteristics in the AUC curve. Higher AUC values denote the better efficiency of the model to detect deepfakes.

Apart from accuracy and AUC, other evaluation metrics are exploited to prove model effectiveness and robustness of the detection model namely, specificity, sensitivity/recall, precision, f1-score, and confusion matrix. Specificity computes the ratio number of true negatives detected with respect to actual number of negative samples in the dataset. Sensitivity determines the ratio of the number of true positives detected with respect to the actual number of positives present in the datasets. On the other hand, precision defined the ratio of number of true positives detected by the algorithm to the total number of samples predicted by the modal. F1-score computes the harmonic mean of precision and recall for computing the detection performance of the algorithms. The confusion matrix diagrammatically represents the similarities between the misclassified classes of the samples which supports to analysis of the areas that need to be considered for improved detection accuracy. These evaluation metrics are an indicator of model performance for both single-modal and multi-modal deepfake detection algorithms before the real-time deployment.

7.4. A short note on ethical implications and societal impact of deepfakes detection

The realistic generation of deepfake content has raised ethical concerns related to the privacy and security of the individual. The contents can impersonate an individual, distrust, and impact its public reputation. These contents have a negative societal impact which may harm many industrial applications such as film production, political campaigns, video games, and many others. Hence, it is crucial to design robust, accurate, and efficient deepfake detection algorithms to prevent the ill impacts of deepfakes from penetrating society.

There exists a wide range of real-time examples where deepfake contents were created and circulated online to spread disinformation. Politics is one of the key areas where deepfake contents are circulated to discredit the image of politicians and leaders. One of the most popular deepfake videos represented the conversation between three American leaders Barak Obama, Donald Trump, and George W. Bush [148]. These individuals appeared to be talking with each other with accurate lip movements. However, no such conversation had

happened among these three ever. Another potentially manipulated deepfake video was circulated in which Barack Obama can be seen insulting Donald Trump [148]. Recently, a prominent deepfake campaign ad featuring the manipulated voice of Kamala Harris has escalated debate over the misuse of the technology [149].

There is an exponential increase in crime against women using deepfakes. Most of the pornographic content featuring females is generated using deepfakes [148]. These explicit videos unethically present the renowned and famous celebrities without their consent. These contents harass people sexually, mentally, and socially. Other notable crimes that can be committed using deepfakes are writing blackmail emails, sending phishing emails, and many other illegal activities. The dominance of highly reliable, accurate, and efficient deepfake detection algorithms is required to prohibit the bad effects of deepfakes on society.

7.5. Recommendations and suggestions for future directions

Based on the selected deepfake detection studies under each category, recommendations are being clustered to provide future insights into the upcoming research. Despite of availability of highly trained DL and transformer networks, the detection of highly realistic synthetic multimedia content is a challenge. The recommendations are (a) Robust system design (b) Diverse and real-time datasets (c) AI architectural vision and (d) detection in low-quality videos. The discussion of these recommendations is as follows (a) Robust system design: The system performance is degraded when the short-term features are extracted which are not suitable to maintain model performance on unseen data [27]. Hence, there is a requirement for a system that can extract long-term robust and discriminative features from the considered deepfake contents. The success rate can be improved by capturing the fake pixels from the long-term temporal and contextual features. (b) Diverse and real-time datasets: Most of the available benchmark datasets consist of synthetic deepfake images for model training. The size of these datasets is not sufficiently large and the number of fake and real images with diversified facial expressions is also limited. Real-time scenarios consisting of both audio and video are not captured properly for tuning the model. (c) AI architectural vision: Various DL-models are adopted for feature extraction and classification. These models are highly complex with a sufficiently large number of convolutional layers, pooling layers, and complicated loss and activation functions [65]. The training of these models requires large datasets and high computational modeling. Hence, the model training and processing should be analyzed to evaluate its capabilities for real-time predictions. Also, the model architecture should be efficient enough in terms of computations and processing. (d) Detection in low-quality videos: To reduce the storage requirements, the video images are highly compressed, and detection of deepfakes in such content is extremely tedious [66]. The deepfake detection model should be trained in such a way so that fake content can also be predicted in low-quality and poor-resolution videos. Detecting deepfakes in well-generated realistic synthetic content is a potential challenge in current scenarios.

Nowadays deepfakes creation is quite easy, quick, and inexpensive [150]. Most of the deepfake creation tools are freely available, and open source and do not require any technical expertise for their operations and usage. The outcomes of these tools are quite fast and very realistic which is very tedious for the existing detection algorithms to identify. Most of the advanced deepfake detection tools have shown superior results in deepfake identification but their performance is limited in practical scenarios. There is a requirement for strong policies and countermeasures to prevent the negative impact of deepfakes on society. Datasets and detection tools must be standardized to design and develop reliable detection algorithms. Generally, the negative impact of deepfakes has prevailed through online platforms [20]. There is a need to design strict, convincing, and robust governing policies to stop their online propagation. The liability of social media and online platforms in terms of their responsibility and obligations should be regulated to prevent the dissemination of false information.

There is a requirement for strict and flexible rules to mitigate the negative impacts of deepfakes on the community and society [58]. The detection algorithms must be equipped with the latest AI-based technologies to identify forgery quickly and in real time. The deepfake detection algorithms should be improved in terms of generalizability and explainability. These features are critical to assess the detection algorithm's performance on cross-datasets and unseen contents. The human intervention for detecting deepfakes is valuable to address the practical detection challenges. Embedding ethical, societal, and community aspects in deepfake detection is significantly important to mitigate its adverse impacts on humans.

Further, the future requirements demand to protect the people from the dangerous impact of deepfakes. For this, it is essential to transform the deepfake algorithms into user-friendly applications and software. This application will be platform friendly which can be integrated easily with social media platforms which can prevent users from deceiving by false content. In addition, the software can be open source which can also promote real-time detection of deepfakes.

8. Conclusion

In this study, we have extensively summarized the various deepfakes detection frameworks under two categories namely, single-modal and multi-modal approaches. It has been analyzed that some contents are manipulated using deepfake creation techniques for fun and comic purposes. But generative AI advancements have created content for negative purposes that include generating pornographic material from popular celebs using face-swap, and augmenting audio of political leaders are few generic examples. These contents are harmful to society and nation, and hence, detection is essential to prevent the dangerous influence. In this study, we have categorized the deepfake detection frameworks based on the number of input features, methodology, fusion strategy, and classification module. Conventional deepfake detection algorithms are widely used due to low-level feature extraction and simple classification modules. On the other hand, deep learning and hybrid deepfake detection approaches utilize a lot of training data for evaluating model performance. On the other hand, multi-modal frameworks provide more realistic scenarios by using multiple modalities in the feature extraction module and fusing them to generate unified features for efficient deepfake classification. Models generalizability is computed and tested using cross-dataset evaluation. In addition, explainable models are also designed for better interpretable outcomes.

CRediT authorship contribution statement

Ashish Kumar: Writing – original draft, Methodology, Investigation, Conceptualization. **Divya Singh:** Writing – review & editing,. **Rachna Jain:** Writing – review & editing. **Deepak Kumar Jain:** Writing – review & editing, Validation, Supervision. **Chenquan Gan:** Writing – review & editing, Supervision. **Xudong Zhao:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] Y. Patel, S. Tanwar, R. Gupta, P. Bhattacharya, I.E. Davidson, R. Nyameko, S. Aluvala, V. Vimal, Deepfake generation and detection: Case study and challenges, IEEE Access 11 (2023) 143296–143323, http://dx.doi.org/10.1109/ACCESS.2023.3342107.

[2] J.W. Seow, M.K. Lim, R.C. Phan, J.K. Liu, A comprehensive overview of deepfake: Generation, detection, datasets, and opportunities, Neurocomputing 513 (2022) 351–371.

[3] K. Somoray, D.J. Miller, Providing detection strategies to improve human detection of deepfakes: An experimental study, Comput. Hum. Behav. 149 (2023) 107917.

[4] T.T. Nguyen, Q.V.H. Nguyen, D.T. Nguyen, D.T. Nguyen, T. Huynh-The, S. Nahavandi, C.M. Nguyen, Deep learning for deepfakes creation and detection: A survey, Comput. Vis. Image Underst. 223 (2022) 103525.

[5] H.R. Hasan, K. Salah, R. Jayaraman, I. Yaqoob, M. Omar, NFTs for combating deepfakes and fake metaverse digital contents, Internet Things 25 (2024) 101133.

[6] B. Gowrisankar, V.L. Thing, An adversarial attack approach for explainable AI evaluation on deepfake detection models, Comput. Secur. 139 (2024) 103684.

[7] M. Mcuba, A. Singh, R.A. Ikuesan, H. Venter, The effect of deep learning methods on deepfake audio detection for digital investigation, Procedia Comput. Sci. 219 (2023) 211–219.

[8] P.M.G.I. Reis, R.O. Ribeiro, A forensic evaluation method for DeepFake detection using DCNN-based facial similarity scores, Forensic Sci. Int. 358 (2024) 111747.

[9] V.V.V.N.S. Vamsi, S.S. Shet, S.S.M. Reddy, S.S. Rose, S.R. Shetty, S. Sathvika, S.P. Shankar, Deepfake detection in digital media forensics, Glob. Transitions Proc. 3 (1) (2022) 74–79.

[10] Z. Xia, T. Qiao, M. Xu, N. Zheng, S. Xie, Towards deepfake video forensics based on facial textural disparities in multi-color channels, Inform. Sci. 607 (2022) 654–669.

[11] A. Kumar, Visual Object Tracking Using Deep Learning, CRC Press, 2023.

[12] A. Kumar, P. Sarren, Raja, Deep learning-based multi-object tracking, in: Object Tracking Technology: Trends, Challenges and Applications, Springer, 2023, pp. 183–199.

[13] A. Kumar, R. Vohra, R. Jain, M. Li, C. Gan, D.K. Jain, Correlation filter based single object tracking: A review, Inf. Fusion (2024) 102562.

[14] A. Kumar, G.S. Walia, K. Sharma, A novel approach for multi-cue feature fusion for robust object tracking, Appl. Intell. 50 (10) (2020) 3201–3218.

[15] A. Kumar, G.S. Walia, K. Sharma, Robust object tracking based on adaptive multicue feature fusion, J. Electron. Imaging 29 (6) (2020) 063001–063001.

[16] A. Kumar, V. Aelgani, R. Vohra, S.K. Gupta, M. Bhagawati, S. Paul, J.R. Laird, Artificial intelligence bias in medical system designs: A systematic review, Multimedia Tools Appl. 83 (6) (2024) 18005–18057.

[17] A. Kumar, R. Jain, Behavioral prediction of cancer using machine learning, in: Cancer Prediction for Industrial IoT 4.0, Chapman and Hall/CRC, 2021, pp. 91–105.

[18] A. Kumar, R. Jain, M. Gupta, S.M. Islam, 6G-Enabled IoT and AI for Smart Healthcare: Challenges, Impact, and Analysis, CRC Press, 2023.

[19] A. Kumar, P. Sareen, A. Arora, Healthcare engineering using AI and distributed technologies, in: Smart Distributed Embedded Systems for Healthcare Applications, CRC Press, 2023, pp. 1–14.

[20] F. Abbas, A. Taeihagh, Unmasking deepfakes: A systematic review of deepfake detection and generation techniques using artificial intelligence, Expert Syst. Appl. (2024) 124260.

[21] A. Mitra, S.P. Mohanty, P. Corcoran, E. Kougianos, A novel machine learning based method for deepfake video detection in social media, in: 2020 IEEE International Symposium on Smart Electronic Systems (ISES) (Formerly INiS), 2020, pp. 91–96, http://dx.doi.org/10.1109/iSES50453.2020.00031.

[22] A. Mitra, S.P. Mohanty, P. Corcoran, E. Kougianos, A machine learning based approach for deepfake detection in social media through key video frame extraction, SN Comput. Sci. 2 (2) (2021) 98.

[23] M.S. Rana, B. Murali, A.H. Sung, Deepfake detection using machine learning algorithms, in: 2021 10th International Congress on Advanced Applied Informatics, IIAI-AAI, IEEE, 2021, pp. 458–463.

[24] A. Qadir, R. Mahum, M.A. El-Meligy, A.E. Ragab, A. AlSalman, M. Awais, An efficient deepfake video detection using robust deep learning, Heliyon 10 (5) (2024).

[25] P. Sadhana, N. Ravishankar, A. Ashok, R. Ravichandran, R. Paul, K. Murali, Enhancing fake image detection: A novel two-step approach combining GANs and CNNs, Procedia Comput. Sci. 235 (2024) 810–819.

[26] S. Suganthi, M.U.A. Ayoobkhan, N. Bacanin, K. Venkatachalam, H. Štěpán, T. Pavel, Deep learning model for deep fake face recognition and detection, PeerJ Comput. Sci. 8 (2022).

[27] Y. Zhu, C. Zhang, J. Gao, X. Sun, Z. Rui, X. Zhou, High-compressed deepfake video detection with contrastive spatiotemporal distillation, Neurocomputing 565 (2024) 126872.

[28] A. Hamza, A.R.R. Javed, F. Iqbal, N. Kryvinska, A.S. Almadhor, Z. Jalil, R. Borghol, Deepfake audio detection via MFCC features using machine learning, IEEE Access 10 (2022) 134018–134028.

[29] M. Tian, M. Khayatkhoei, J. Mathai, W. AbdAlmageed, Unsupervised multi-modal deepfake detection using intra-and cross-modal inconsistencies, 2024, arXiv preprint arXiv:2311.17088.

[30] F. Iqbal, A. Abbasi, A.R. Javed, Z. Jalil, J.N. Al-Karaki, Deepfake audio detection via feature engineering and machine learning, in: CIKM Workshops, 2022, pp. 1–12.

[31] G. Wang, Q. Jiang, X. Jin, X. Cui, FFR_FD: Effective and fast detection of DeepFakes via feature point defects, Inform. Sci. 596 (2022) 472–488.

[32] X. Yang, Y. Li, S. Lyu, Exposing deep fakes using inconsistent head poses, in: ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2019, pp. 8261–8265.

[33] H. Farid, Creating, using, misusing, and detecting deep fakes, J. Online Trust. Saf. 1 (4) (2022).

[34] F. Khalid, A. Javed, H. Ilyas, A. Irtaza, DFGNN: An interpretable and generalized graph neural network for deepfakes detection, Expert Syst. Appl. 222 (2023) 119843.

[35] X. Nguyen, T. Tran, K. Nguyen, D.-T. Truong, Learning spatio-temporal features to detect manipulated facial videos created by the deepfake techniques, Forensic Sci. Int.: Digit. Investig. 36 (2021) 301108.

[36] W. Pu, J. Hu, X. Wang, Y. Li, S. Hu, B. Zhu, S. Lyu, Learning a deep dual-level network for robust DeepFake detection, Pattern Recognit. 130 (2022) 108832.

[37] S. Dincer, G. Ulutas, B. Ustubioglu, G. Tahaoglu, N. Sklavos, Golden ratio based deep fake video detection system with fusion of capsule networks, Comput. Electr. Eng. 117 (2024) 109234.

[38] E. Essa, Feature fusion vision transformers using MLP-mixer for enhanced deepfake detection, Neurocomputing 598 (2024) 128128.

[39] S. Ganguly, A. Ganguly, S. Mohiuddin, S. Malakar, R. Sarkar, ViXNet: Vision transformer with xception network for deepfakes based video and image forgery detection, Expert Syst. Appl. 210 (2022) 118423.

[40] D. Coccomini, R. Caldelli, F. Falchi, C. Gennaro, On the generalization of deep learning models in video deepfake detection, J. Imaging 9 (5) (2023) 89.

[41] X. Liu, Y. Yu, X. Li, Y. Zhao, Mcl: multimodal contrastive learning for deepfake detection, IEEE Trans. Circuits Syst. Video Technol. (2024).

[42] J. Gao, M. Micheletto, G. Orrù, S. Concas, X. Feng, G. Marcialis, F. Roli, Texture and artifact decomposition for improving generalization in deep-learning-based deepfake detection, Eng. Appl. Artif. Intell. 133 (2024) 108450.

[43] S. Kingra, N. Aggarwal, N. Kaur, LBPNet: Exploiting texture descriptor for deepfake detection, Forensic Sci. Int.: Digit. Investig. 42 (2022) 301452.

[44] G. Leporoni, L. Maiano, L. Papa, I. Amerini, A guided-based approach for deepfake detection: RGB-depth integration via features fusion, Pattern Recognit. Lett. 181 (2024) 99–105.

[45] A. Elhassan, M. Al-Fawa'reh, M. Jafar, M. Ababneh, S. Jafar, DFT-MF: Enhanced deepfake detection using mouth movement and transfer learning, SoftwareX 19 (2022) 101115.

[46] R. Saravana Ram, M. Vinoth Kumar, T. Al-shami, M. Masud, H. Aljuaid, M. Abouhawwash, Deep fake detection using computer vision-based deep neural network with pairwise learning, Intell. Autom. Soft Comput. 35 (2) (2023).

[47] X. Xu, S. Tang, M. Zhu, P. He, S. Li, Y. Cao, A novel model compression method based on joint distillation for deepfake video detection, J. King Saud Univ. Comput. Inf. Sci. 35 (9) (2023) 101792.

[48] Z. Xu, J. Liu, W. Lu, B. Xu, X. Zhao, B. Li, J. Huang, Detecting facial manipulated videos based on set convolutional neural networks, J. Vis. Commun. Image Represent. 77 (2021) 103119.

[49] M.A. Raza, K.M. Malik, Multimodaltrace: Deepfake detection using audiovisual representation learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 993–1000.

[50] J. Yoon, A. Panizo-LLedot, D. Camacho, C. Choi, Triple-modality interaction for deepfake detection on zero-shot identity, Inf. Fusion 109 (2024) 102424.

[51] S. Muppalla, S. Jia, S. Lyu, Integrating audio-visual features for multimodal deepfake detection, in: 2023 IEEE MIT Undergraduate Research Technology Conference, URTC, IEEE, 2023, pp. 1–5.

[52] D. Salvi, H. Liu, S. Mandelli, P. Bestagini, W. Zhou, W. Zhang, S. Tubaro, A robust approach to multimodal deepfake detection, J. Imaging 9 (6) (2023) 122.

[53] Z. Yan, P. Sun, Y. Lang, S. Du, S. Zhang, W. Wang, L. Liu, Multimodal graph learning for deepfake detection, 2022, arXiv preprint arXiv:2209.05419.

[54] M. Rana, M. Nobi, B. Murali, A. Sung, Deepfake detection: A systematic literature review, IEEE Access 10 (2022) 25494–25513.

[55] L. Passos, D. Jodas, K. Costa, L. Souza Júnior, D. Rodrigues, J. Del Ser, J. Papa, A review of deep learning-based approaches for deepfake content detection, Expert Syst. 41 (8) (2024) e13570.

[56] M. Albahar, J. Almalki, Deepfakes: Threats and countermeasures systematic review, J. Theor. Appl. Inf. Technol. 97 (22) (2019) 3242–3250.

[57] A. Malik, M. Kuribayashi, S.M. Abdullahi, A.N. Khan, DeepFake detection for human face images and videos: A survey, IEEE Access 10 (2022) 18757–18775.

[58] G. Gupta, K. Raja, M. Gupta, T. Jan, S.T. Whiteside, M. Prasad, A comprehensive review of DeepFake detection using advanced machine learning and fusion methods, Electronics 13 (1) (2023) 95.

[59] A. Mary, A. Edison, Deep fake detection using deep learning techniques: A literature review, in: 2023 International Conference on Control, Communication and Computing, ICCC, IEEE, 2023, pp. 1–6.

[60] R. Tolosana, S. Romero-Tapiador, R. Vera-Rodriguez, E. Gonzalez-Sosa, J. Fierrez, DeepFakes detection across generations: Analysis of facial regions, fusion, and performance evaluation, Eng. Appl. Artif. Intell. 110 (2022) 104673.

[61] B. Yan, C.-T. Li, X. Lu, JRC: Deepfake detection via joint reconstruction and classification, Neurocomputing (2024) 127862.

[62] A. Ismail, M. Elpeltagy, M. S. Zaki, K. Eldahshan, A new deep learning-based methodology for video deepfake detection using XGBoost, Sensors 21 (16) (2021) 5413.

[63] A. Raza, K. Munir, M. Almutairi, A novel deep learning approach for deepfake image detection, Appl. Sci. 12 (19) (2022) 9820.

[64] A. Kharel, M. Paranjape, A. Bera, DF-TransFusion: Multimodal deepfake detection via lip-audio cross-attention and facial self-attention, 2023, arXiv preprint arXiv:2309.06511.

[65] D. Wodajo, S. Atnafu, Deepfake video detection using convolutional vision transformer, 2021, arXiv preprint arXiv:2102.11126.

[66] M.A. Amin, Y. Hu, C.-T. Li, B. Liu, Deepfake detection based on cross-domain local characteristic analysis with multi-domain transformer, Alex. Eng. J. 91 (2024) 592–609.

[67] J. Khochare, C. Joshi, B. Yenarkar, S. Suratkar, F. Kazi, A deep learning framework for audio deepfake detection, Arab. J. Sci. Eng. (2021) 1–12.

[68] G. Naskar, S. Mohiuddin, S. Malakar, E. Cuevas, R. Sarkar, Deepfake detection using deep feature stacking and meta-learning, Heliyon 10 (4) (2024).

[69] D.S. AbdElminaam, N. Sherif, Z. Ayman, M. Mohamed, M. Hazem, DeepFakeDG: A deep learning approach for deep fake detection and generation, J. Comput. Commun. 2 (2) (2023) 31–37.

[70] S.T. Ikram, S. Chambial, D. Sood, et al., A performance enhancement of deepfake video detection through the use of a hybrid CNN deep learning model, Int. J. Electr. Comput. Eng. Syst. 14 (2) (2023) 169–178.

[71] L. Guarnera, O. Giudice, S. Battiato, Deepfake detection by analyzing convolutional traces, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops, 2020, pp. 666–667.

[72] M. Kim, S. Tariq, S.S. Woo, Fretal: Generalizing deepfake detection using knowledge distillation and representation learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2021, pp. 1001–1012.

[73] A.A. Maksutov, V.O. Morozov, A.A. Lavrenov, A.S. Smirnov, Methods of deepfake detection based on machine learning, in: 2020 IEEE Conference of Russian Young Researchers in Electrical and Electronic Engineering, EIConRus, IEEE, 2020, pp. 408–411.

[74] H. Zou, M. Shen, Y. Hu, C. Chen, E.S. Chng, D. Rajan, Cross-modality and within-modality regularization for audio-visual deepfake detection, in: ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2024, pp. 4900–4904.

[75] O. De Lima, S. Franklin, S. Basu, B. Karwoski, A. George, Deepfake detection using spatiotemporal convolutional networks, 2020, arXiv preprint arXiv:2006.14749.

[76] R. Liu, J. Zhang, G. Gao, Multi-space channel representation learning for mono-to-binaural conversion based audio deepfake detection, Inf. Fusion 105 (2024) 102257.

[77] Y. Nirkin, L. Wolf, Y. Keller, T. Hassner, Deepfake detection based on discrepancies between faces and their context, IEEE Trans. Pattern Anal. Mach. Intell. 44 (10) (2021) 6111–6121.

[78] B. Wang, X. Wu, F. Wang, Y. Zhang, F. Wei, Z. Song, Spatial-frequency feature fusion based deepfake detection through knowledge distillation, Eng. Appl. Artif. Intell. 133 (2024) 108341.

[79] L.-Y. Hsu, AI-assisted deepfake detection using adaptive blind image watermarking, J. Vis. Commun. Image Represent. 100 (2024) 104094.

[80] K. Jayashre, M. Amsaprabhaa, Safeguarding media integrity: A hybrid optimized deep feature fusion based deepfake detection in videos, Comput. Secur. 142 (2024) 103860.

[81] G. Lee, M. Kim, Deepfake detection using the rate of change between frames based on computer vision, Sensors 21 (21) (2021) 7367.

[82] J.K. Lewis, I.E. Toubal, H. Chen, V. Sandesera, M. Lomnitz, Z. Hampel-Arias, C. Prasad, K. Palaniappan, Deepfake video detection based on spatial, spectral, and temporal inconsistencies using multimodal deep learning, in: 2020 IEEE Applied Imagery Pattern Recognition Workshop, AIPR, IEEE, 2020, pp. 1–9.

[83] H. Ilyas, A. Javed, K. Malik, AvFakeNet: A unified end-to-end dense swin transformer deep learning model for audio–visual deepfakes detection, Appl. Soft Comput. 136 (2023) 110124.

[84] C. Zhu, B. Zhang, Q. Yin, C. Yin, W. Lu, Deepfake detection via inter-frame inconsistency recomposition and enhancement, Pattern Recognit. 147 (2024) 110077.

[85] Y. Ni, W. Zeng, P. Xia, G. Yang, R. Tan, A deepfake detection algorithm based on Fourier transform of biological signal, Comput. Mater. Contin. 79 (3) (2024).

[86] M.S. Rana, A.H. Sung, Deepfakestack: A deep ensemble-based learning technique for deepfake detection, in: 2020 7th IEEE International Conference on Cyber Security and Cloud Computing (CSCloud)/2020 6th IEEE International Conference on Edge Computing and Scalable Cloud, EdgeCom, IEEE, 2020, pp. 70–75.

[87] N. Yu, L. Chen, T. Leng, Z. Chen, X. Yi, An explainable deepfake of speech detection method with spectrograms and waveforms, J. Inf. Secur. Appl. 81 (2024) 103720.

[88] M. Arshed, S. Mumtaz, M. Ibrahim, C. Dewi, M. Tanveer, S. Ahmed, Multiclass AI-generated deepfake face detection using patch-wise deep learning model, Computers 13 (1) (2024) 31.

[89] Z. Guo, G. Yang, J. Chen, X. Sun, Fake face detection via adaptive manipulation traces extraction network, Comput. Vis. Image Underst. 204 (2021) 103170.

[90] F. Nie, J. Ni, J. Zhang, B. Zhang, W. Zhang, FRADE: Forgery-aware audio-distilled multimodal learning for deepfake detection, in: ACM Multimedia 2024, 2024, pp. 1–10.

[91] L. Zhang, D. Zhao, C. Lim, H. Asadi, H. Huang, Y. Yu, R. Gao, Video deepfake classification using particle swarm optimization-based evolving ensemble models, Knowl.-Based Syst. 289 (2024) 111461.

[92] D. Ballesteros, Y. Rodriguez-Ortega, D. Renza, G. Arce, Deep4SNet: deep learning for fake speech classification, Expert Syst. Appl. 184 (2021) 115465.

[93] M. Lomnitz, Z. Hampel-Arias, V. Sandesara, S. Hu, Multimodal approach for deepfake detection, in: 2020 IEEE Applied Imagery Pattern Recognition Workshop, AIPR, IEEE, 2020, pp. 1–9.

[94] S. Solaiyappan, Y. Wen, Machine learning based medical image deepfake detection: A comparative study, Mach. Learn. Appl. 8 (2022) 100298.

[95] J. Awotunde, R. Jimoh, A. Imoize, A. Abdulrazaq, C.-T. Li, C.-C. Lee, An enhanced deep learning-based deepfake video detection and classification system, Electronics 12 (1) (2022) 87.

[96] A. Badale, L. Castelino, C. Darekar, J. Gomes, Deep fake detection using neural networks, in: 15th IEEE International Conference on Advanced Video and Signal Based Surveillance, AVSS, 2018, pp. 1–6.

[97] M. Raza, K. Malik, I. Haq, Holisticdfd: Infusing spatiotemporal transformer embeddings for deepfake detection, Inform. Sci. 645 (2023) 119352.

[98] P. Sharma, M. Kumar, H. Sharma, GAN-CNN ensemble: A robust deepfake detection model of social media images using minimized catastrophic forgetting and generative replay technique, Procedia Comput. Sci. 235 (2024) 948–960.

[99] I. Haq, K. Malik, K. Muhammad, Multimodal neurosymbolic approach for explainable deepfake detection, ACM Trans. Multimed. Comput. Commun. Appl. (2023).

[100] R. Rafique, M. Nawaz, H. Kibriya, M. Masood, Deepfake detection using error level analysis and deep learning, in: 2021 4th International Conference on Computing & Information Sciences, ICCIS, IEEE, 2021, pp. 1–4.

[101] C. Ledig, L. Theis, F. Huszár, J. Caballero, A. Cunningham, A. Acosta, Z. Wang, Photo-realistic single image super-resolution using a generative adversarial network, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2017).

[102] Z. He, W. Zuo, M. Kan, S. Shan, X. Chen, Attgan: Facial attribute editing by only changing what you want, IEEE Trans. Image Process. 28 (11) (2019) 5464–5478.

[103] T. Karras, S. Laine, T. Aila, A style-based generator architecture for generative adversarial networks, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019.

[104] P.K. R., R. Mukhopadhyay, A. Jha, V. Namboodiri, C. Jawahar, Towards automatic face-to-face translation, in: Proceedings of the 27th ACM International Conference on Multimedia, 2019.

[105] R. Natsume, T. Yatagawa, S. Morishima, Rsgan: face swapping and editing using face and hair representation in latent spaces, 2018, arXiv preprint arXiv:1804.03447.

[106] Z. Khanjani, G. Watson, V. Janeja, How deep are the fakes? focusing on audio deepfake: A survey, 2021, arXiv preprint arXiv:2111.14203.

[107] A. Khodabakhsh, C. Busch, R. Ramachandra, A taxonomy of audiovisual fake multimedia content creation technology, in: 2018 IEEE Conference on Multimedia Information Processing and Retrieval, MIPR, IEEE, 2018.

[108] S. Salman, J. Shamsi, R. Qureshi, Deep fake generation and detection: Issues, challenges, and solutions, IT Prof. 25 (1) (2023) 52–59.

[109] K. Liu, I. Perov, D. Gao, N. Chervoniy, W. Zhou, W. Zhang, Deepfacelab: Integrated, flexible and extensible face-swapping framework, Pattern Recognit. 141 (2023) 109628.

[110] Y. Wang, L. Song, W. Wu, C. Qian, R. He, C. Loy, Talking faces: Audio-to-video face generation, in: Handbook of digital face manipulation and detection: From deepfakes to morphing attacks, Springer International Publishing Cham, 2022, pp. 163–188.

[111] J. Thies, M. Zollhofer, M. Stamminger, C. Theobalt, M. Nießner, Face2face: Real-time face capture and reenactment of rgb videos, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 2387–2395.

[112] J. Zhang, X. Zeng, M. Wang, Y. Pan, Y. Liu, Y. Liu, C. Fan, Freenet: Multi-identity face reenactment, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020.

[113] S. Kolagati, T. Priyadharshini, V. Rajam, Exposing deepfakes using a deep multilayer perceptron–convolutional neural network model, Int. J. Inf. Manag. Data Insights 2 (1) (2022) 100054.
[114] H. Lin, W. Huang, W. Luo, W. Lu, DeepFake detection with multi-scale convolution and vision transformer, Digit. Signal Process. 134 (2023) 103895.
[115] Z. Cai, M. Li, Integrating frame-level boundary detection and deepfake detection for locating manipulated regions in partially spoofed audio forgery attacks, Comput. Speech Lang. 85 (2024) 101597.
[116] C.-J. Chew, Y.-C. Lin, Y.-C. Chen, Y.-Y. Fan, J.-S. Lee, Preserving manipulated and synthetic deepfake detection through face texture naturalness, J. Inf. Secur. Appl. 83 (2024) 103798.
[117] J. Ke, L. Wang, DF-UDetector: An effective method towards robust deepfake detection via feature restoration, Neural Netw. 160 (2023) 216–226.
[118] W. Yang, X. Zhou, Z. Chen, B. Guo, Z. Ba, Z. Xia, K. Ren, Avoid-df: Audio-visual joint learning for detecting deepfake, IEEE Trans. Inf. Forensics Secur. 18 (2023) 2015–2029.
[119] H. Khalid, M. Kim, S. Tariq, S.S. Woo, Evaluation of an audio-video multimodal deepfake dataset using unimodal and multimodal detectors, in: Proceedings of the 1st Workshop on Synthetic Multimedia-Audiovisual Deepfake Generation and Detection, 2021, pp. 7–15.
[120] T. Mittal, U. Bhattacharya, R. Chandra, A. Bera, D. Manocha, Emotions don't lie: An audio-visual deepfake detection method using affective cues, in: Proceedings of the 28th ACM International Conference on Multimedia, 2020, pp. 2823–2832.
[121] J. Yang, S. Xiao, A. Li, G. Lan, H. Wang, Detecting fake images by identifying potential texture difference, Future Gener. Comput. Syst. 125 (2021) 127–135.
[122] S.-Y. Lim, D.-K. Chae, S.-C. Lee, Detecting deepfake voice using explainable deep learning techniques, Appl. Sci. 12 (8) (2022) 3926.
[123] D. Cozzolino, A. Pianese, M. Nießner, L. Verdoliva, Audio-visual person-of-interest deepfake detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 943–952.
[124] C. Liy, L. InIctuOculi, Exposing AI-created fake videos by detecting eye blinking, in: 2018 IEEE International Workshop on Information Forensics and Security, WIFS, IEEE, 2018.
[125] P. Korshunov, S. Marcel, D. Fakes, A New Threat to Face Recognition? Assessment and Detection, Cornell University, New York, NY, USA, 2018, pp. 1–5.
[126] A. Rossler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, M. Nießner, Faceforensics++: Learning to detect manipulated facial images, Proc. IEEE/ CVF Int. Conf. Comput. Vis. (2019).
[127] B. Dolhansky, The deepfake detection challenge (DFDC) pre view dataset, 2019, arXiv preprint arXiv:1910.08854.
[128] Y. Li, X. Yang, P. Sun, H. Qi, S. Lyu, Celeb-DF: A large-scale challenging dataset for deepfake forensics, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020.
[129] L. Jiang, R. Li, W. Wu, C. Qian, C. Loy, Deeperforensics-1.0: A large-scale dataset for real-world face forgery detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020.
[130] B. Zi, M. Chang, J. Chen, X. Ma, Y.-G. Jiang, Wilddeepfake: A challenging real-world dataset for deepfake detection, in: Proceedings of the 28th ACM international conference on multimedia, 2020.
[131] H. Dang, F. Liu, J. Stehouwer, X. Liu, A. Jain, On the detection of digital face manipulation, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020.
[132] T.-N. Le, H.H. Nguyen, J. Yamagishi, I. Echizen, Openforensics: Large-scale challenging dataset for multi-face forgery detection and segmentation in-the-wild, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10117–10127.
[133] A. Rössler, D. Cozzolino, L. Verdoliva, C. Riess, J. Thies, M. Nießner, Faceforensics: A large-scale video dataset for forgery detection in human faces, 2018, arXiv preprint arXiv:1803.09179.
[134] J. Yamagishi, M. Todisco, M. Sahidullah, H. Delgado, X. Wang, N. Evans, T. Kinnunen, K.A. Lee, V. Vestman, A. Nautsch, Asvspoof 2019: The 3rd automatic speaker verification spoofing and countermeasures challenge database, 2019.
[135] J. Kominek, A. Black, The CMU arctic speech databases, in: Fifth ISCA Workshop on Speech Synthesis, 2004, pp. 1–2.
[136] K. Park, T. Mulc, CSS10: A collection of single speaker speech datasets for 10 languages, 2019, arXiv preprint arXiv:1903.11269.
[137] K. MacLean, Voxforge, 2018, http://www.voxforge.org/home, (Accessed on 2012).
[138] Z. Cai, S. Ghosh, A. Dhall, T. Gedeon, K. Stefanov, M. Hayat, Glitch in the matrix: A large scale benchmark for content driven audio–visual forgery detection and localization, Comput. Vis. Image Underst. 236 (2023) 103818.
[139] M. Taeb, H. Chi, Comparison of deepfake detection techniques through deep learning, J. Cybersecur. Priv. 2 (1) (2022) 89–106.
[140] M. Albahar, J. Almalki, Deepfakes: Threats and countermeasures systematic review, J. Theor. Appl. Inf. Technol. 97 (22) (2019) 3242–3250.
[141] A. Kumar, R. Mamgai, R. Jain, Application of IoT-enabled CNN for natural language processing, in: IoT-Enabled Convolutional Neural Networks: Techniques and Applications, River Publishers, 2023, pp. 149–177.
[142] A. Kumar, A. Mangotra, A. Ailawadi, R. Jain, M. Arora, Sentiment analysis on multilingual data: Hinglish, in: International Conference on Data Analytics & Management, Springer, 2023, pp. 607–620.
[143] L. Mohanty, A. Kumar, V. Mehta, M. Agarwal, J. Suri, Pruning techniques for artificial intelligence networks: a deeper look at their engineering design and bias: the first review of its kind, Multimedia Tools Appl. (2024) 1–75.
[144] S. Dong, J. Wang, J. Liang, H. Fan, R. Ji, Explaining deepfake detection by analysing image matching, in: European Conference on Computer Vision, Springer, 2022, pp. 18–35.
[145] Z. Lin, M. Shafiee, S. Bochkarev, M. Jules, X. Wang, A. Wong, Do explanations reflect decisions? A machine-centric strategy to quantify the performance of explainability algorithms, 2019, arXiv preprint arXiv:1910.07387.
[146] W. Samek, G. Montavon, A. Binder, S. Lapuschkin, K.-R. Müller, Interpreting the predictions of complex ml models by layer-wise relevance propagation, 2016, arXiv preprint arXiv:1611.08191.
[147] C. Zhao, C. Wang, G. Hu, H. Chen, C. Liu, J. Tang, ISTVT: interpretable spatial-temporal video transformer for deepfake detection, IEEE Trans. Inf. Forensics Secur. 18 (2023) 1335–1348.
[148] A. Smith, Deepfakes are the most dangerous crime of the future, researchers say, Indep. 5 (2020).
[149] K. Tenbarge, Elon musk made a kamala harris deepfake ad go viral, sparking a debate about parody and free speech, NBCNews (2024) (Retrieved 12 October 2024).
[150] A. Kaur, A. Noori Hoshyar, V. Saikrishna, S. Firmin, F. Xia, Deepfake video detection: challenges and opportunities, Artif. Intell. Rev. 57 (6) (2024) 1–47.